Results — revised submission — original style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	Emotions	593	72	6	1.48	0.16
3	Scene	2407	294	6	1.25	0.11
4	PlantPseAAC	978	440	12	6.69	0.68
5	HumanPseAAC	3106	440	14	15.29	1.05
6	Yeast	2417	103	14	7.20	1.82
7	birds	645	260	19	5.41	0.80
8	medical	978	1449	45	222.37	0.72
9	enron	1702	1001	53	10.45	0.80

Table 1: Datasets used in the revised experiments. N, P, K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **ECC baseline on `birds` (RF + LGBM).** The `birds` folders contain only PA/PR/BR/CC/CLR — ECC has not been run. Method slot 12 currently shows as a gap on every `birds` panel. Re-run with `scripts/train_extra_baselines.py --dataset birds --algorithm ecc` for both `--base_learner rf` and `--base_learner lgbm`.
- **LGBM run on `enron`.** Sequentially training (2–3h per task due to K=53 pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}_{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form <order>-<target>-<heig *order* $\in$ {PA = partial order, PR = pre-order}; *target* $\in$ {H = Hamming, S = Subset 0/1}; *height* $\in$ {2, ∞ (omitted in the label)}.

- 1 = **PA-H-2**: partial order, Hamming target, height ≤ 2.
- 2 = **PA-H**: partial order, Hamming target, unrestricted height.
- 3 = **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2.
- 4 = **PA-S**: partial order, Subset 0/1, unrestricted.
- 5 = **PR-H-2**: pre-order, Hamming, height ≤ 2.
- 6 = **PR-H**: pre-order, Hamming, unrestricted.
- 7 = **PR-S-2**: pre-order, Subset 0/1, height ≤ 2.
- 8 = **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- 9 = **BR** (Binary Relevance): K independent binary classifiers, one per label.
- 10 = **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- 11 = **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- 12 = **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level, colored as in the original paper's Table 5: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$, $\alpha = 0.3$. *PartialAbstention* and *ScoreVector* panels show only indices 1–8 (the remaining baselines do not export a partial-abstention prediction nor a probabilistic score vector). Vertical dashed lines at $x = 4.5$, $x = 8.5$, and $x = 11.5$ separate the four method groups: PA (1–4), PR (5–8), standard baselines BR/CC/CLR (9–11), and ECC (12). Missing methods show as gaps in the line (with a small grey "x" on the axis when the method has no data at all for that panel).

3 Main results: RF base learner, aggregated

RF • Aggregate • BinaryVector — Standard MLC predictions

RF • Aggregate • PartialAbstention — Partial abstention

RF • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

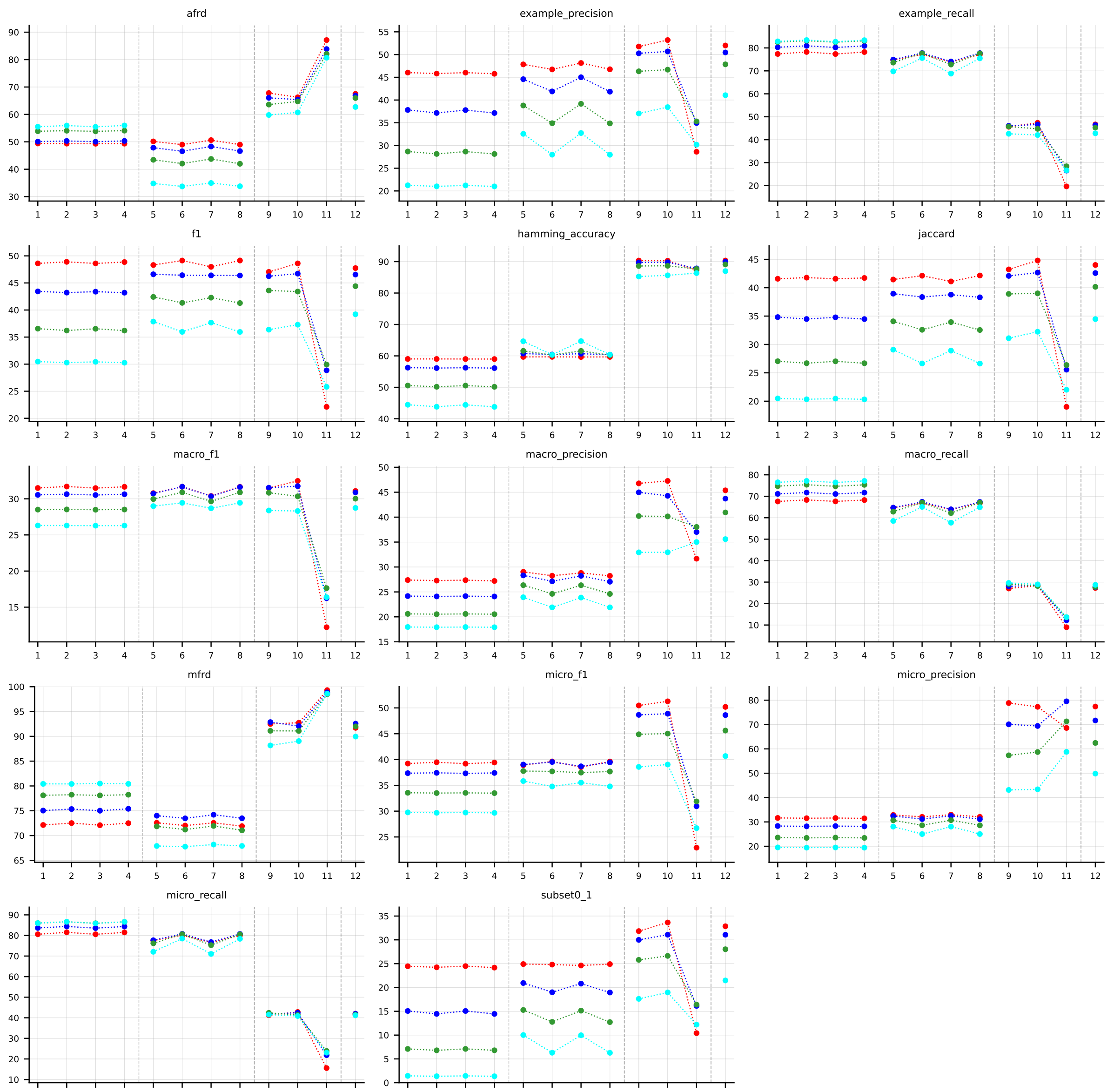

Table 2: Standard MLC predictions (BinaryVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

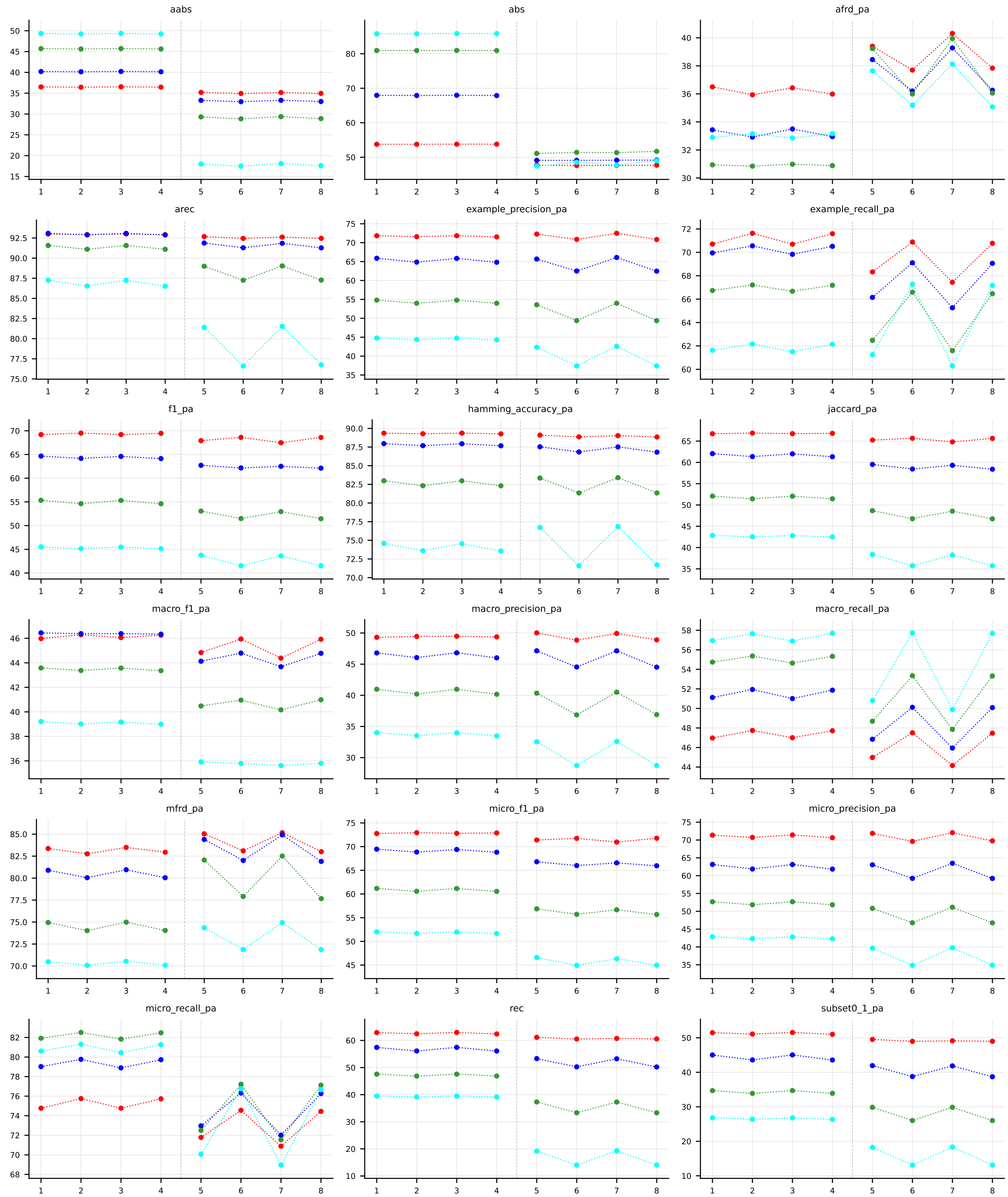

Table 3: Partial abstention (PartialAbstention), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

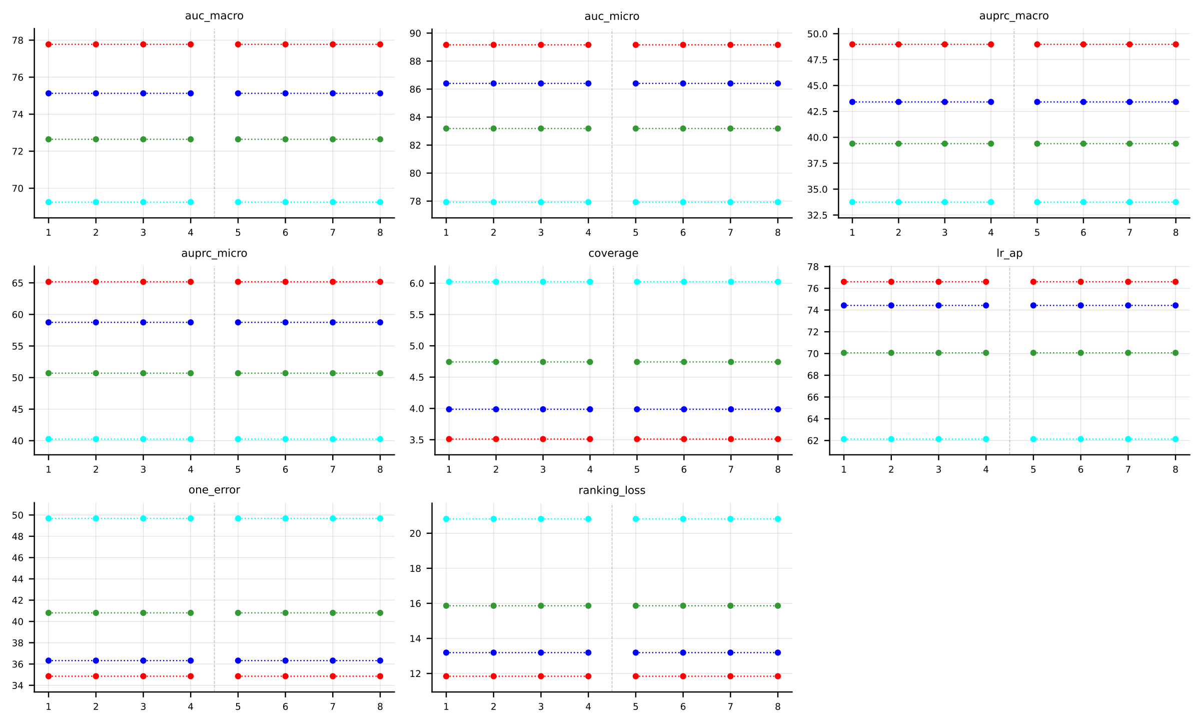

Table 4: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

4 Abstention vs standard MLC

Why this section exists. The paper's central claim is that an order-based predictor can *abstain* on labels it is unsure about and, on the labels it does predict, achieve higher quality than a standard MLC method that is forced to predict every label. The two chart families below make that claim visible.

Terminology. '*Standard MLC*' (a.k.a. `BinaryVector` / BV) means each method outputs a 0/1 prediction for every one of the K labels. '*With abstention*' (a.k.a. `PartialAbstention` / PA) means an order-based predictor may return $\perp$ (abstain) on some labels; the PA quality metrics (`f1_pa`, `jaccard_pa`, ...) are computed only over the non-abstained labels. Only the 8 PA/PR methods (indices 1–8) can abstain; BR/CC/CLR/ECC cannot.

Reading the coverage-risk chart. Each point is one (PA/PR method, noise level α) pair. The x-axis is the abstention rate (fraction of labels the method skipped); the y-axis is the quality metric on the labels that *were* predicted (higher is better). Marker shape encodes method, colour encodes noise level. The dashed grey line is the best standard-MLC baseline (BR / CC / CLR / ECC) on the BV quality metric — it is the bar that "no abstention" has to clear. **Any point above the dashed line is direct evidence that abstaining improved quality beyond what any standard MLC method achieves.** Points further right traded coverage for that gain.

Reading the paired-bars chart. Same data, different view. The four sub-panels correspond to the four noise levels. Inside each sub-panel: 12 method positions, two bars per position — blue is the standard-MLC quality (`f1`, no abstention), warm-colour is the with-abstention quality on retained labels. Only the 8 PA/PR methods (1–8) have a warm bar; the standard MLC baselines (9–12) only have a blue bar by construction. The green "$+x.y$" label above each warm bar is the gain in score-points over that method's own BV bar.

4.1 RF, metric `f1_pa` — aggregate

Figure 1: RF (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.2 RF, metric `f1_pa` — per dataset

Figure 2: RF on `GpositivePseAAC` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 3: RF on `emotions` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 4: RF on `scene` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 5: RF on `PlantPseAAC` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 6: RF on `HumanPseAAC` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 7: RF on `Yeast` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 8: RF on `birds` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 9: RF on `medical` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 10: RF on `enron` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

4.3 RF, metric `jaccard_pa` — aggregate

Figure 11: RF (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.4 RF, metric `jaccard_pa` — per dataset

Figure 12: RF on `GpositivePseAAC` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 13: RF on `emotions` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 14: RF on `scene` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 15: RF on `PlantPseAAC` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 16: RF on `HumanPseAAC` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 17: RF on `Yeast` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 18: RF on `birds` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 19: RF on `medical` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 20: RF on `enron` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

4.5 LGBM, metric `f1_pa` — aggregate

Figure 21: LGBM (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.6 LGBM, metric `f1_pa` — per dataset

Figure 22: LGBM on `GpositivePseAAC` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 23: LGBM on `emotions` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 24: LGBM on `scene` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 25: LGBM on `PlantPseAAC` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 26: LGBM on `HumanPseAAC` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 27: LGBM on `Yeast` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 28: LGBM on `birds` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 29: LGBM on `medical` (f1_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

4.7 LGBM, metric jaccard_pa — aggregate

Figure 30: LGBM (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right), averaged across all datasets.

4.8 LGBM, metric jaccard_pa — per dataset

Figure 31: LGBM on `GpositivePseAAC` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 32: LGBM on `emotions` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 33: LGBM on `scene` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 34: LGBM on `PlantPseAAC` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 35: LGBM on `HumanPseAAC` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 36: LGBM on `Yeast` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 37: LGBM on `birds` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

Figure 38: LGBM on `medical` (jaccard_pa): coverage-risk (left) and standard-MLC-vs-abstention paired bars (right).

A Per-dataset results (RF base learner)

A.1 GpositivePseAAC ($K = 4$)

RF • `GpositivePseAAC` • BinaryVector — Standard MLC predictions

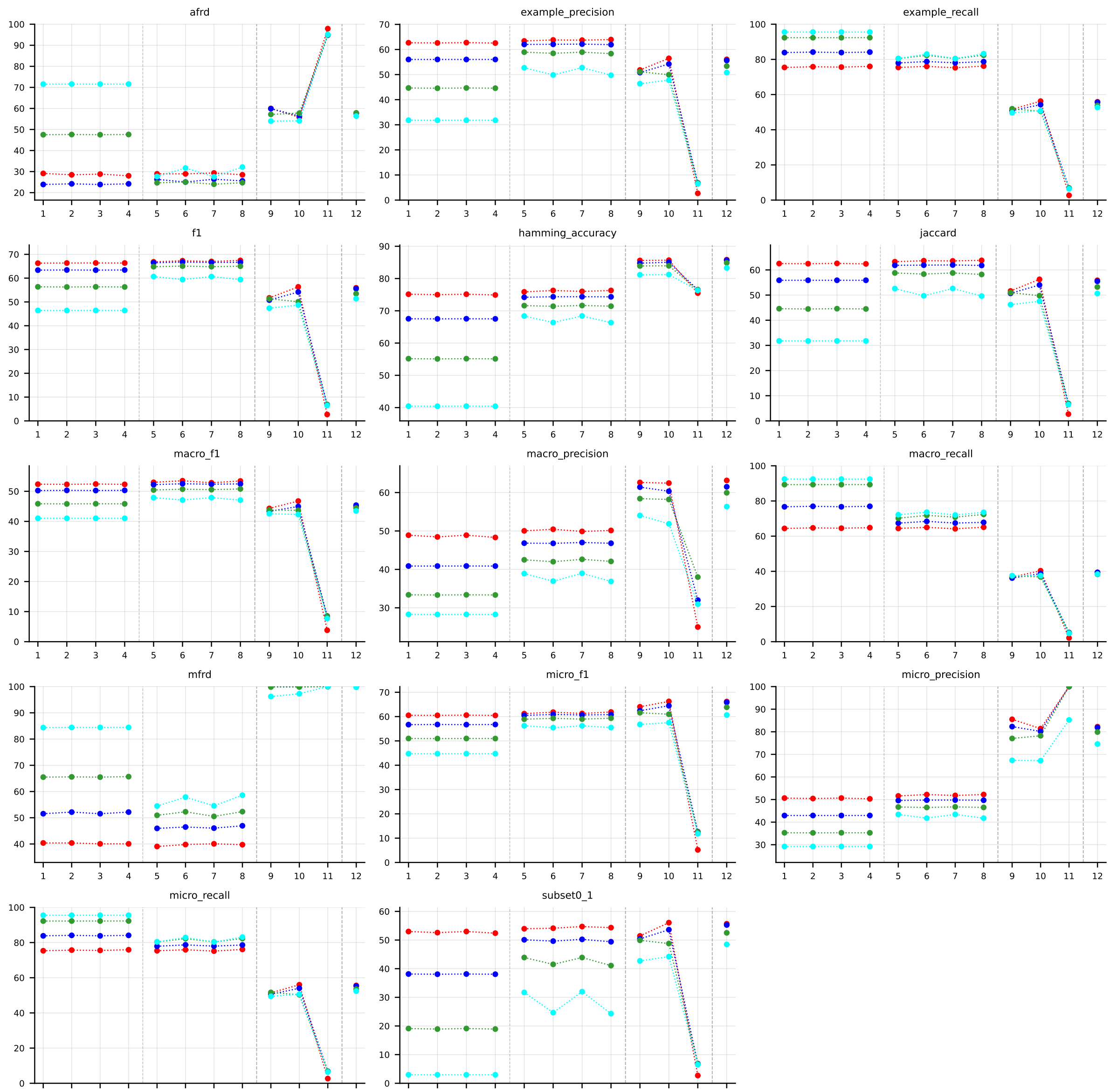

Table 5: Standard MLC predictions (BinaryVector) on `GpositivePseAAC` (RF base learner).

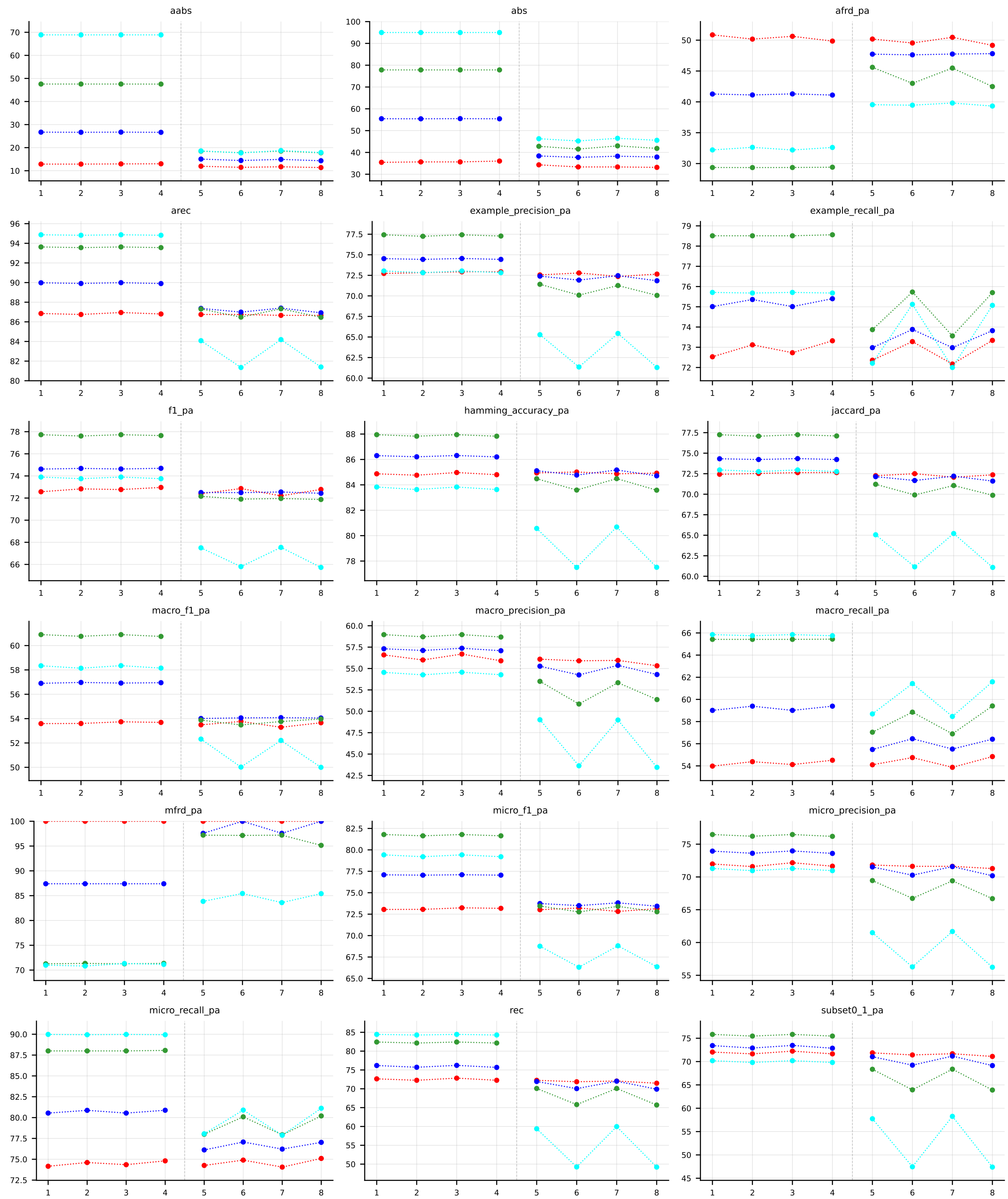

Table 6: Partial abstention (PartialAbstention) on `GpositivePseAAC` (RF base learner).

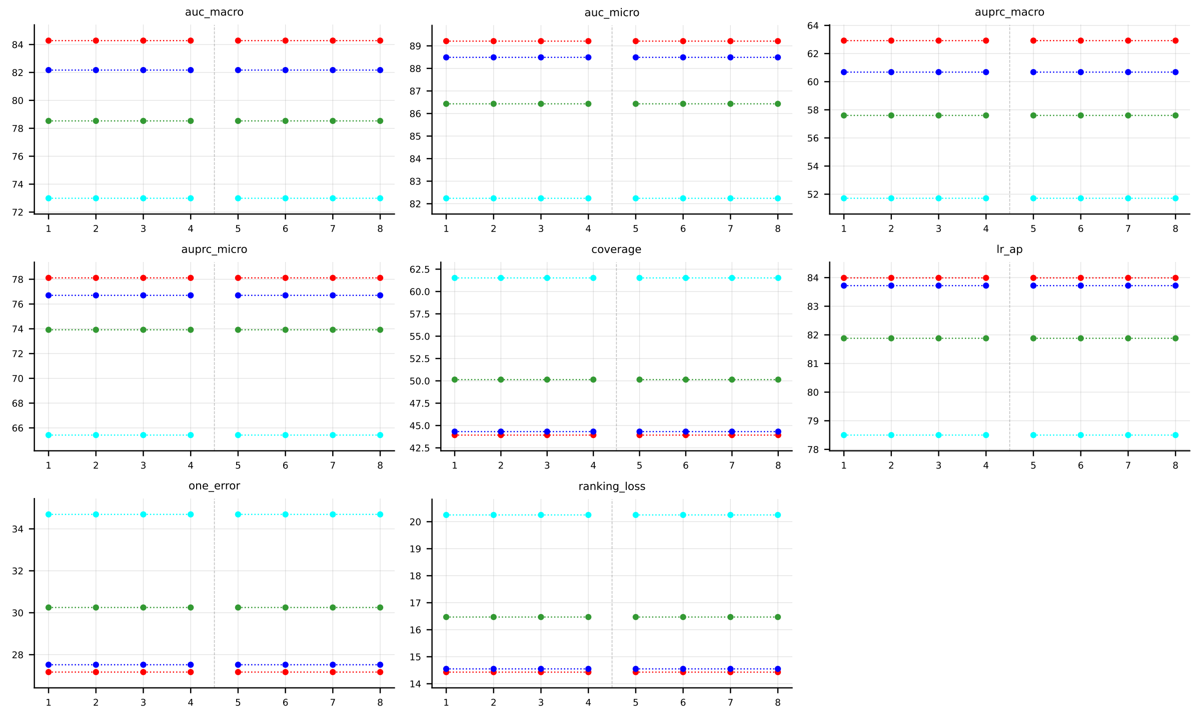

Table 7: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `GpositivePseAAC` (RF base learner).

RF • `GpositivePseAAC` • PartialAbstention — Partial abstention

RF • `GpositivePseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.2 Emotions ($K = 6$)

RF • `emotions` • BinaryVector — Standard MLC predictions

RF • `emotions` • PartialAbstention — Partial abstention

RF • `emotions` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.3 Scene ($K = 6$)

RF • `scene` • BinaryVector — Standard MLC predictions

RF • `scene` • PartialAbstention — Partial abstention

RF • `scene` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.4 PlantPseAAC ($K = 12$)

RF • `PlantPseAAC` • BinaryVector — Standard MLC predictions

RF • `PlantPseAAC` • PartialAbstention — Partial abstention

RF • `PlantPseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.5 HumanPseAAC ($K = 14$)

RF • `HumanPseAAC` • BinaryVector — Standard MLC predictions

RF • `HumanPseAAC` • PartialAbstention — Partial abstention

RF • `HumanPseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

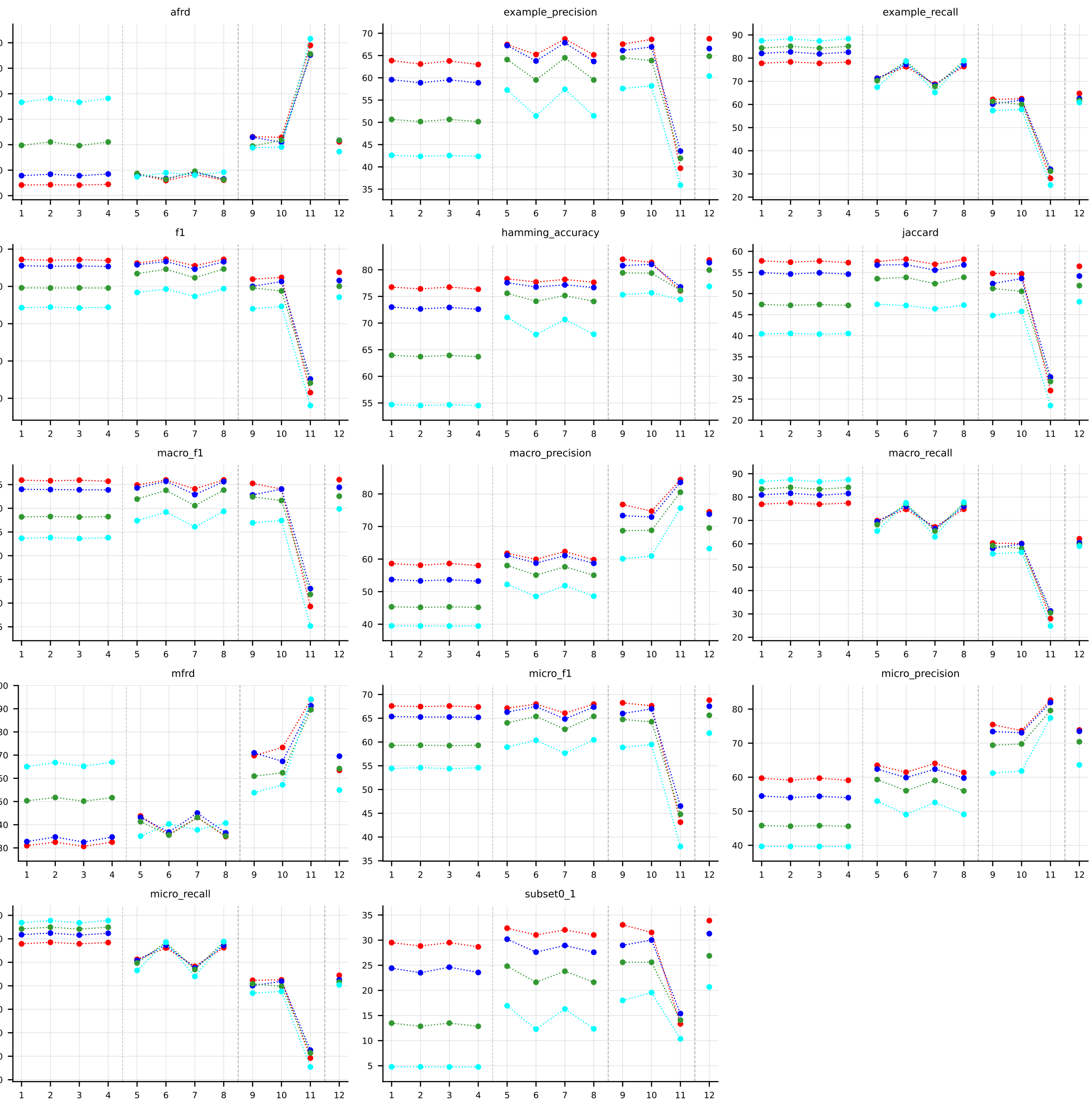

Table 8: Standard MLC predictions (BinaryVector) on `emotions` (RF base learner).

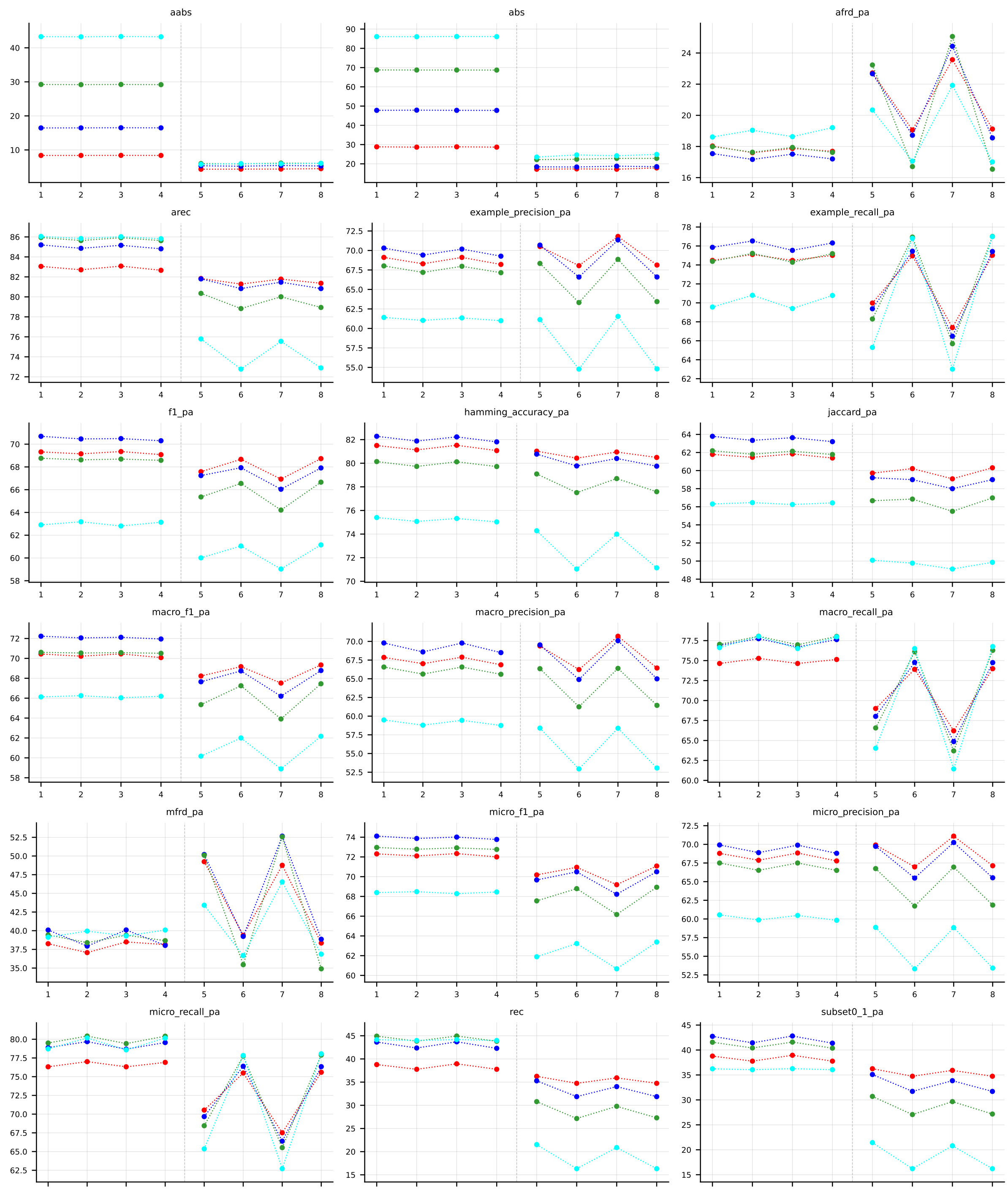

Table 9: Partial abstention (PartialAbstention) on `emotions` (RF base learner).

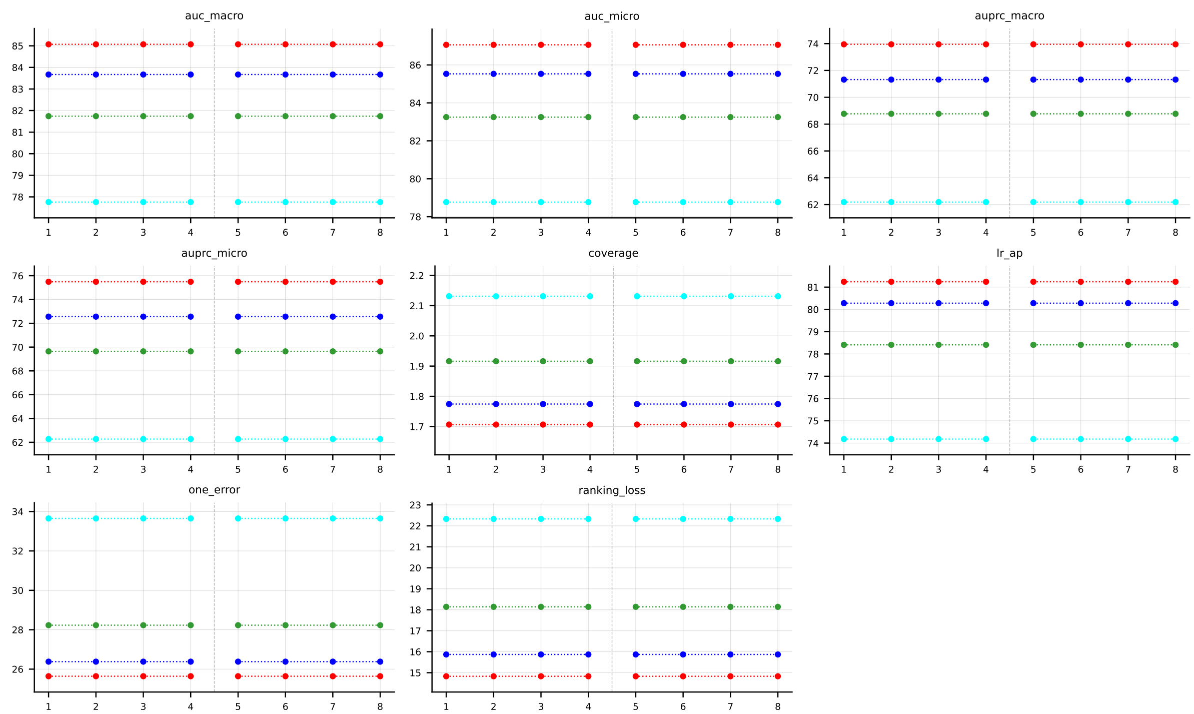

Table 10: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `emotions` (RF base learner).

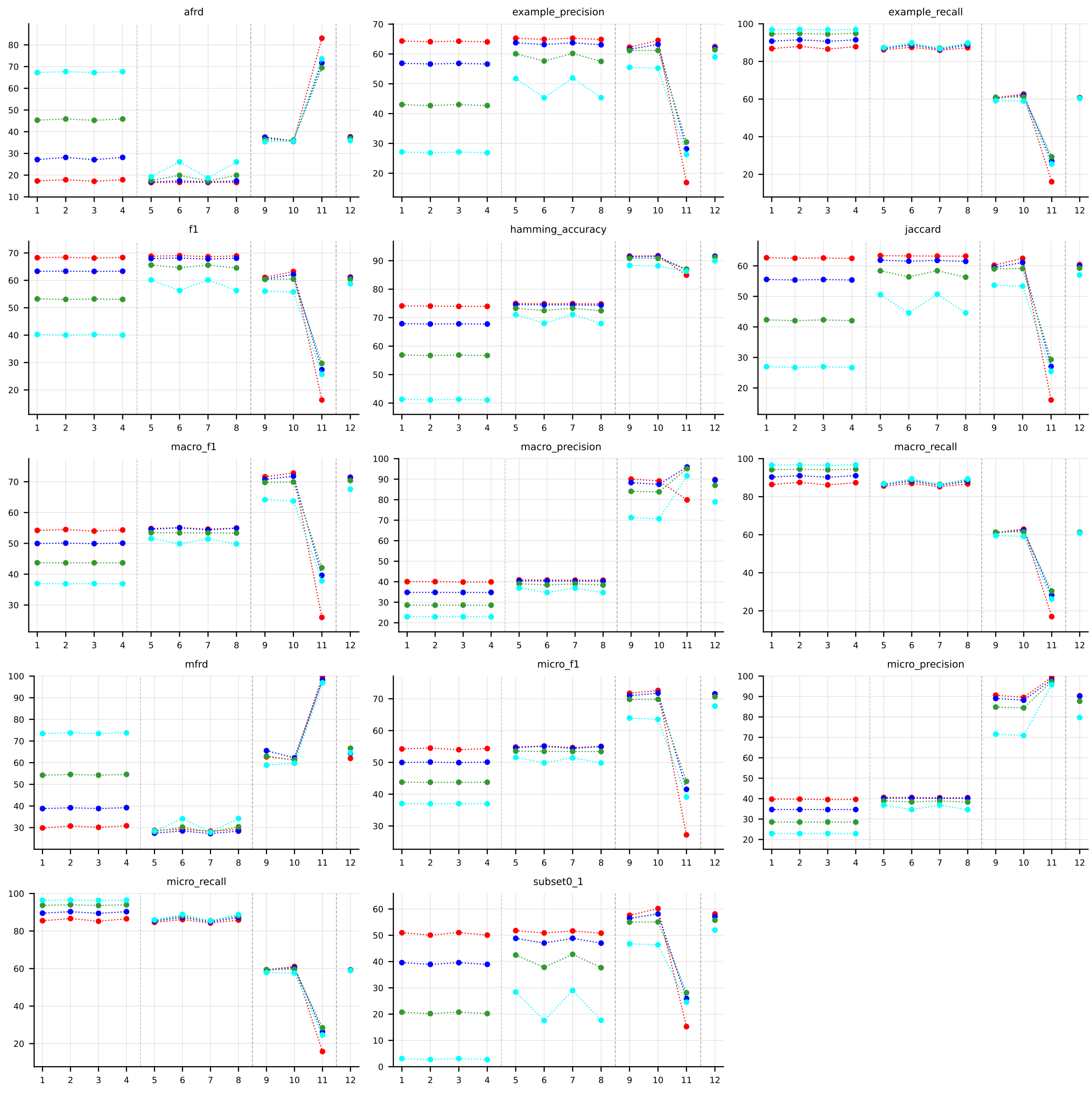

Table 11: Standard MLC predictions (BinaryVector) on `scene` (RF base learner).

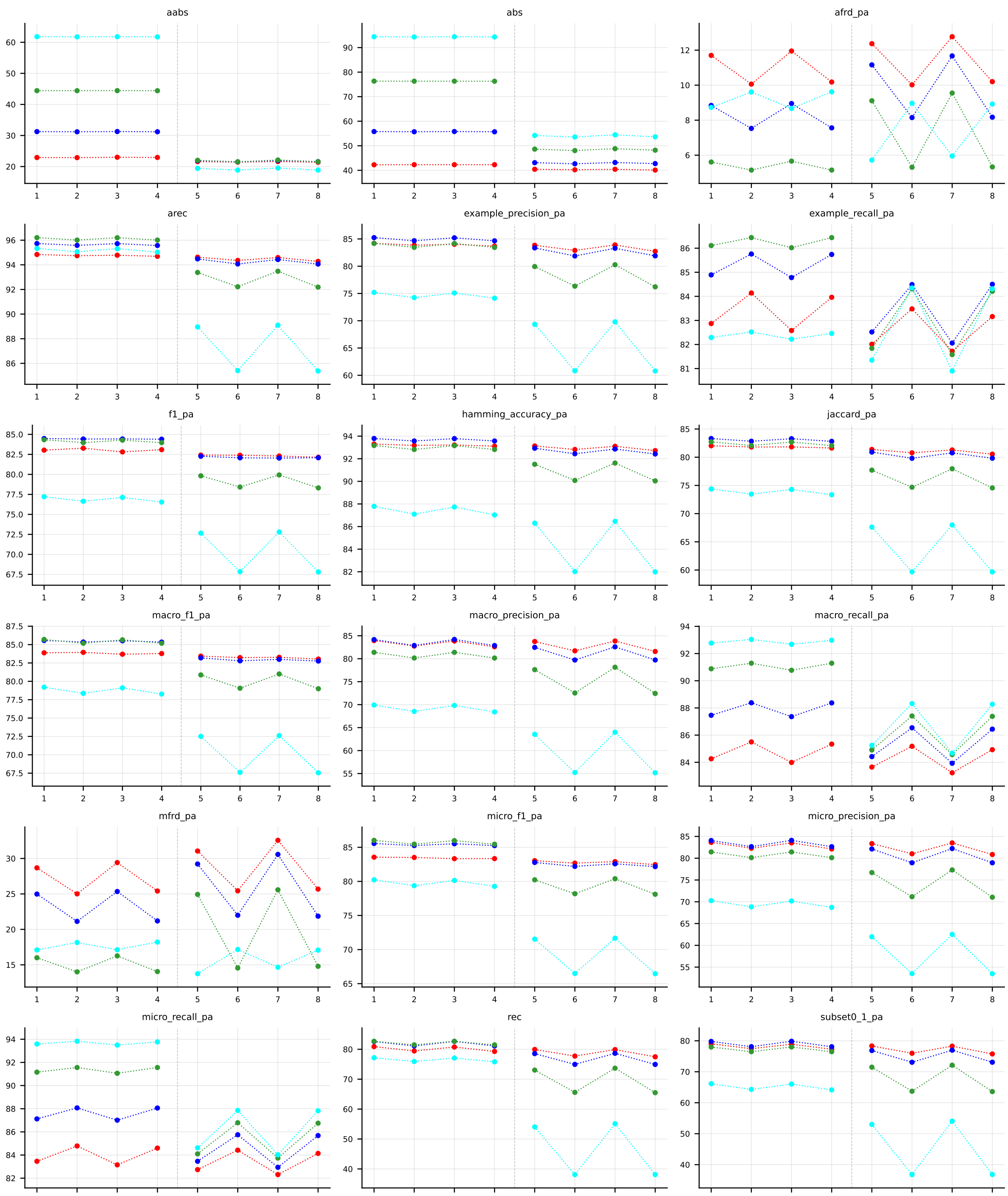

Table 12: Partial abstention (PartialAbstention) on `scene` (RF base learner).

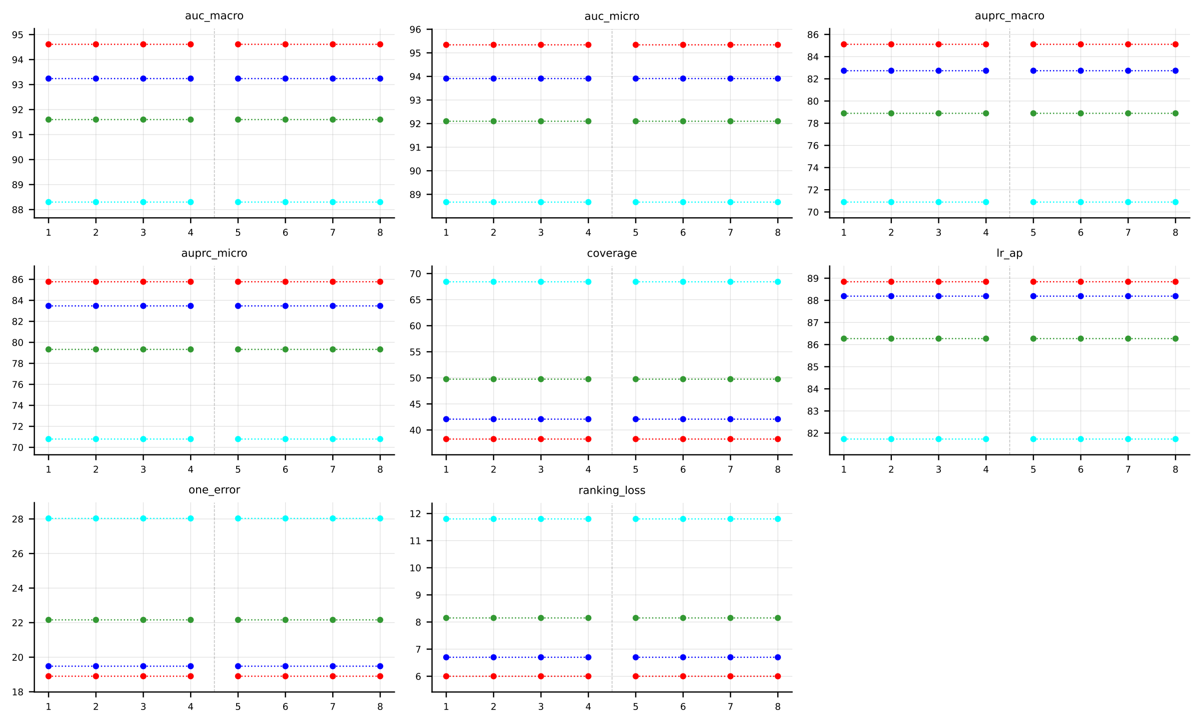

Table 13: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `scene` (RF base learner).

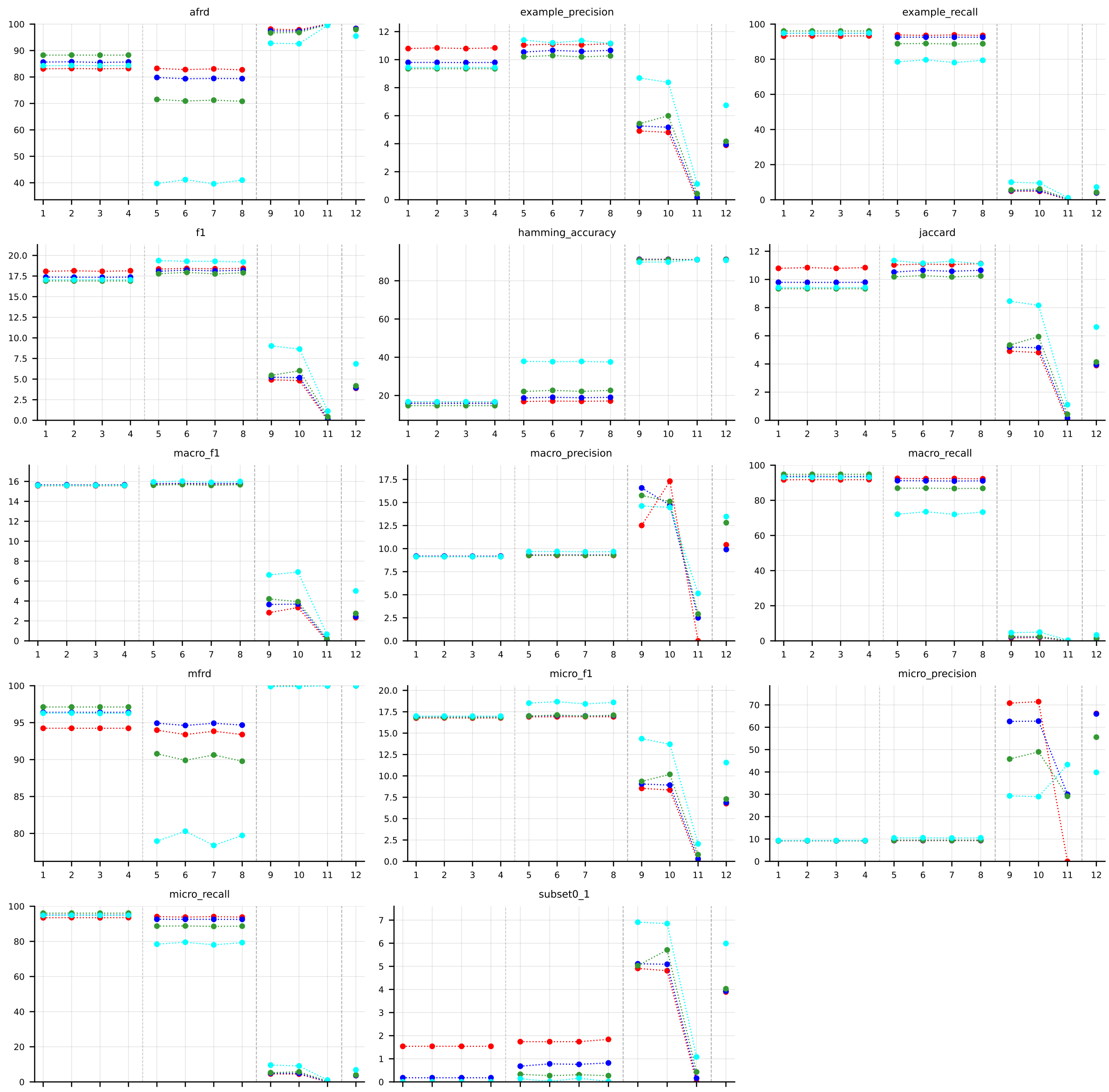

Table 14: Standard MLC predictions (BinaryVector) on `PlantPseAAC` (RF base learner).

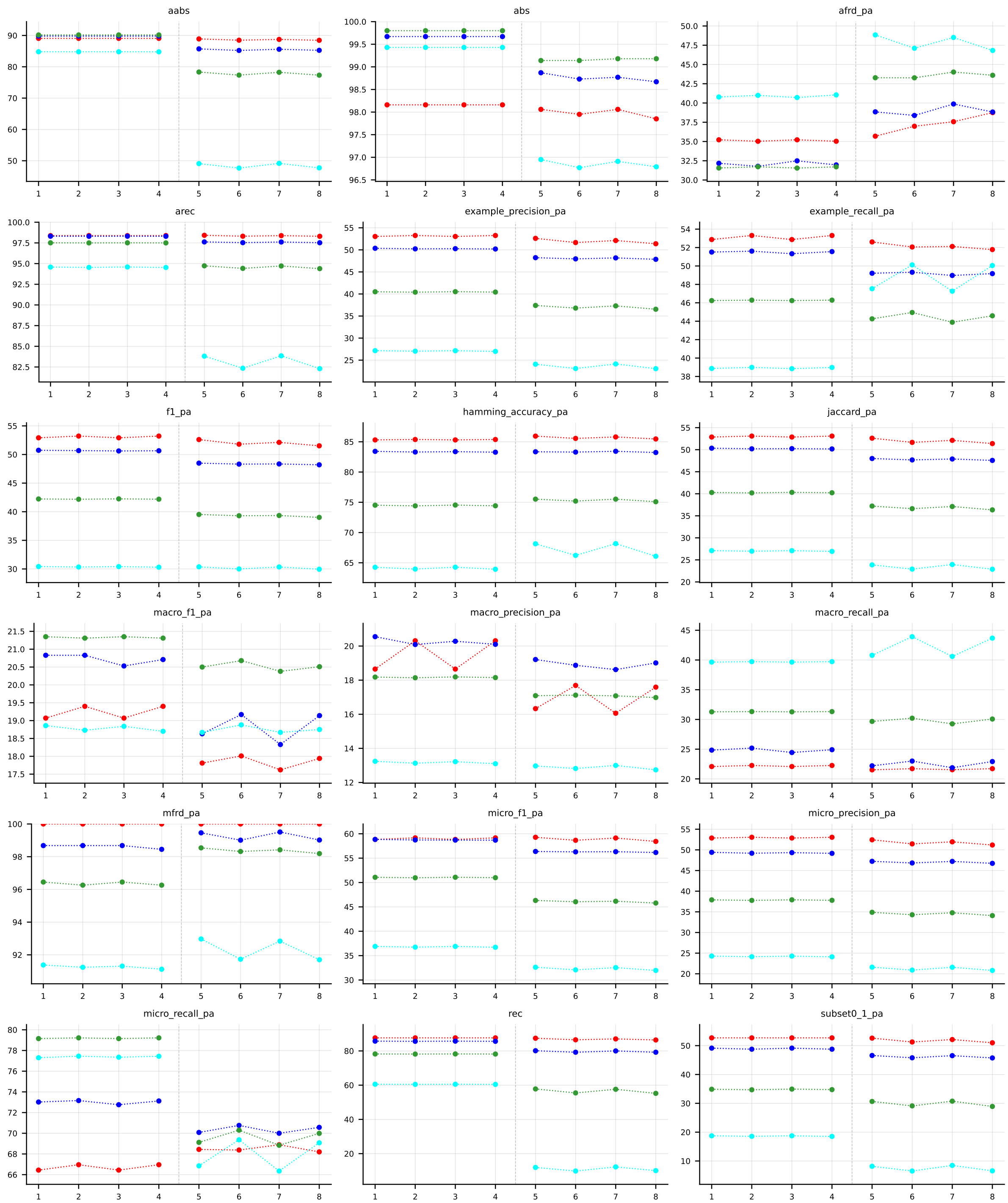

Table 15: Partial abstention (PartialAbstention) on `PlantPseAAC` (RF base learner).

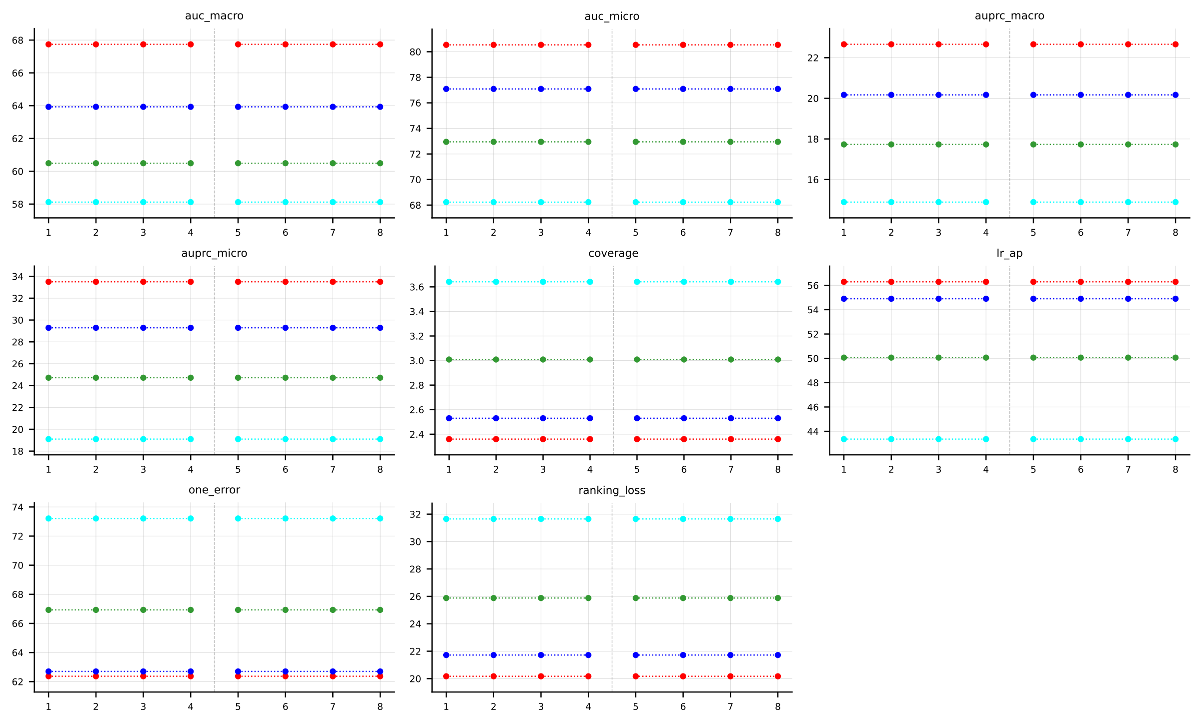

Table 16: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `PlantPseAAC` (RF base learner).

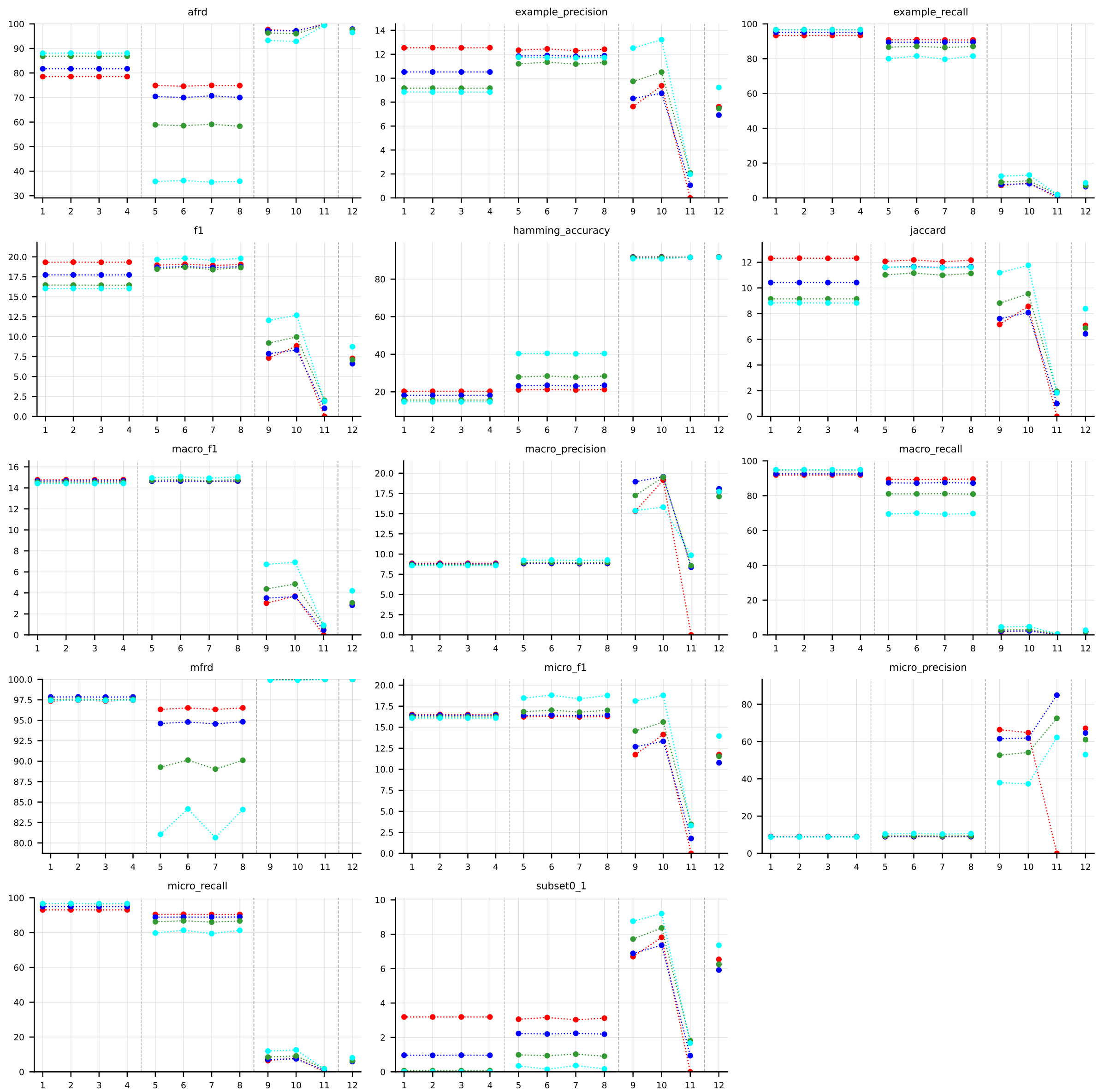

Table 17: Standard MLC predictions (BinaryVector) on `HumanPseAAC` (RF base learner).

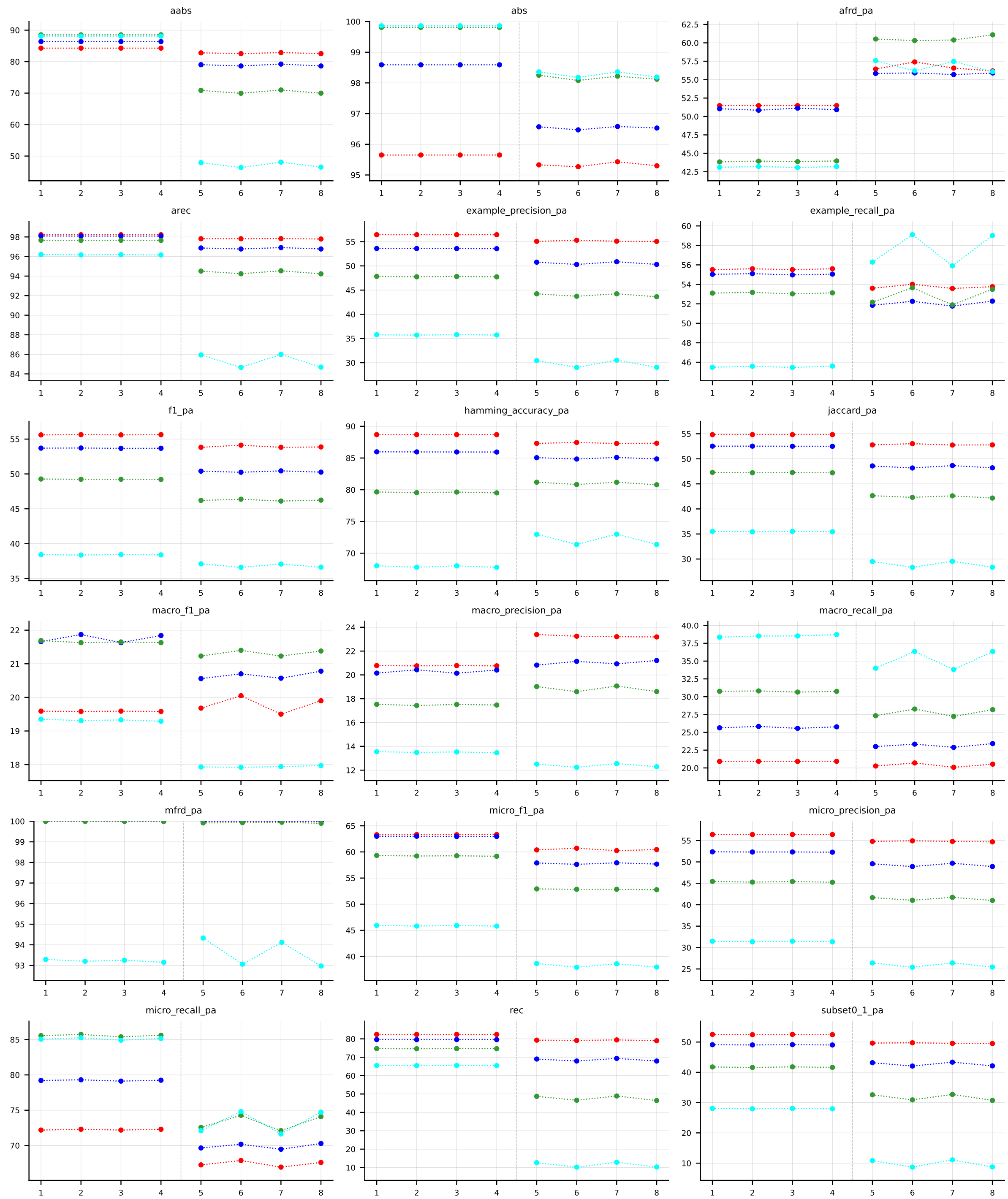

Table 18: Partial abstention (PartialAbstention) on `HumanPseAAC` (RF base learner).

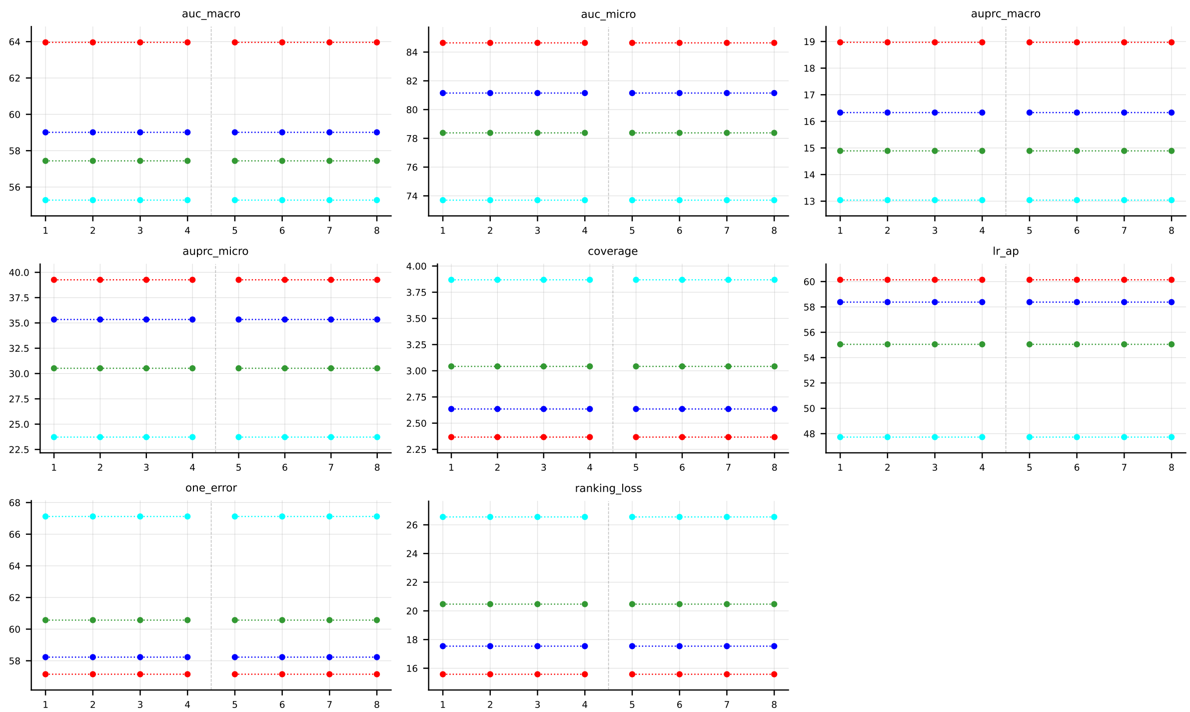

Table 19: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `HumanPseAAC` (RF base learner).

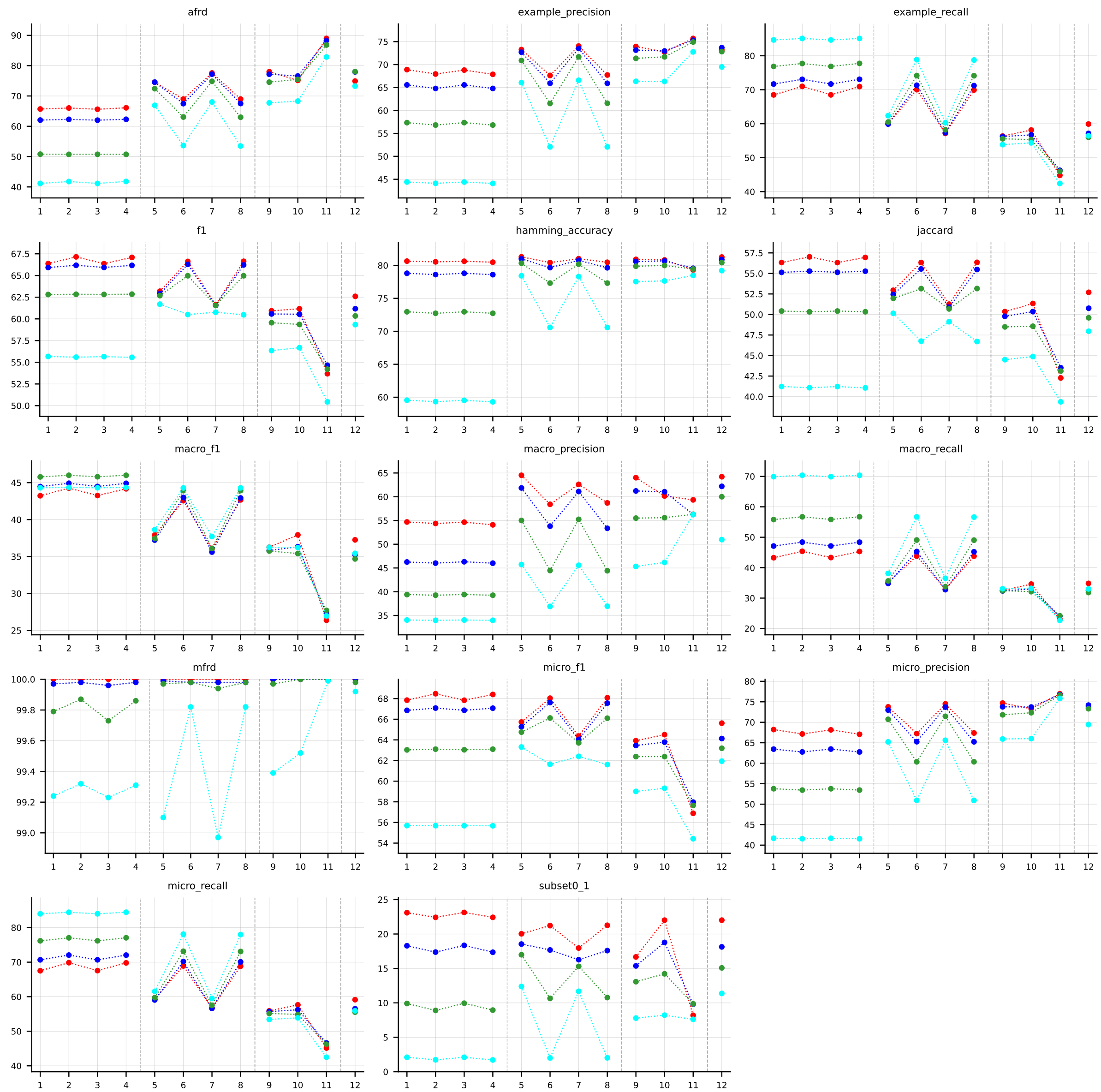

Table 20: Standard MLC predictions (BinaryVector) on `Yeast` (RF base learner).

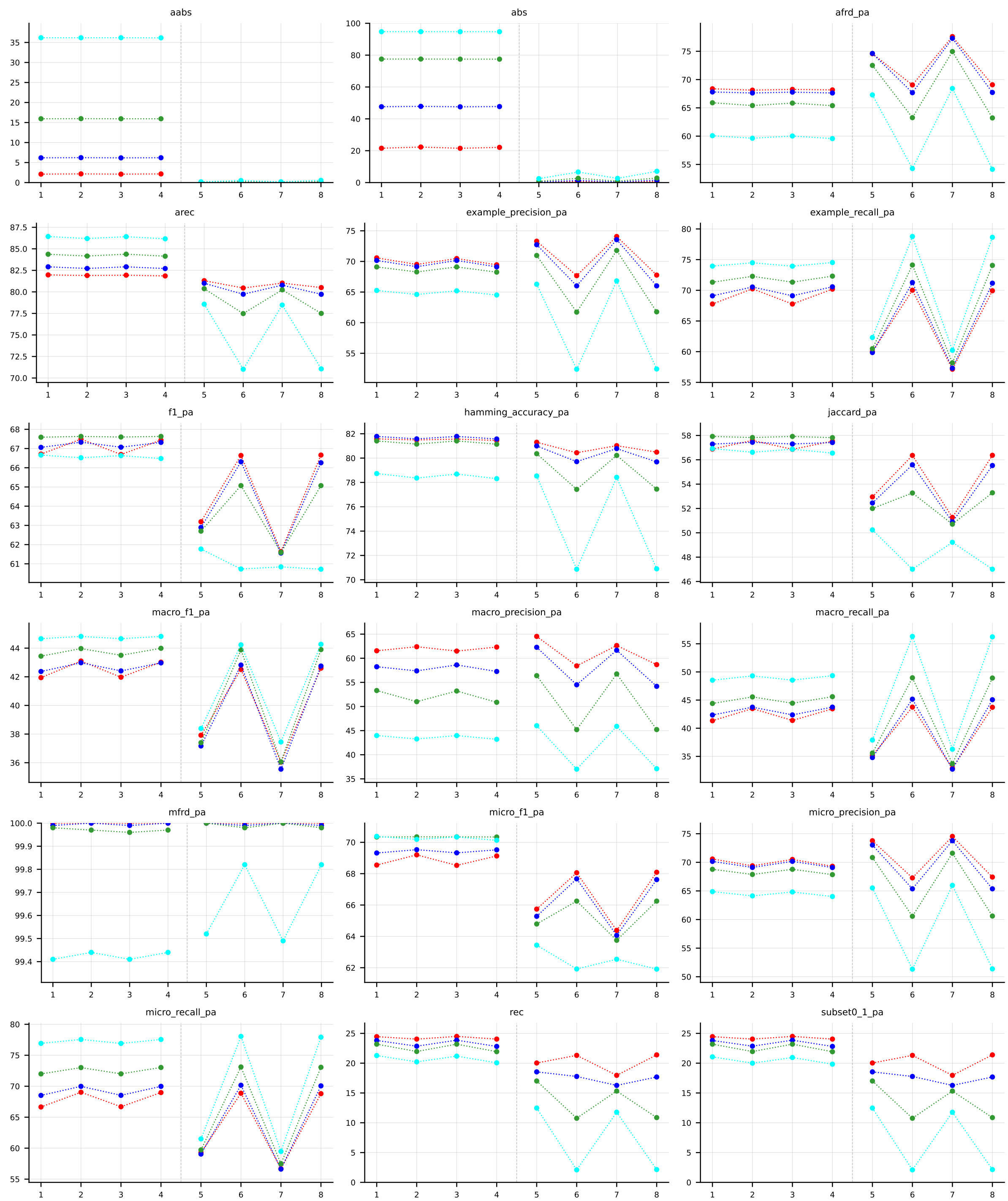

Table 21: Partial abstention (PartialAbstention) on `Yeast` (RF base learner).

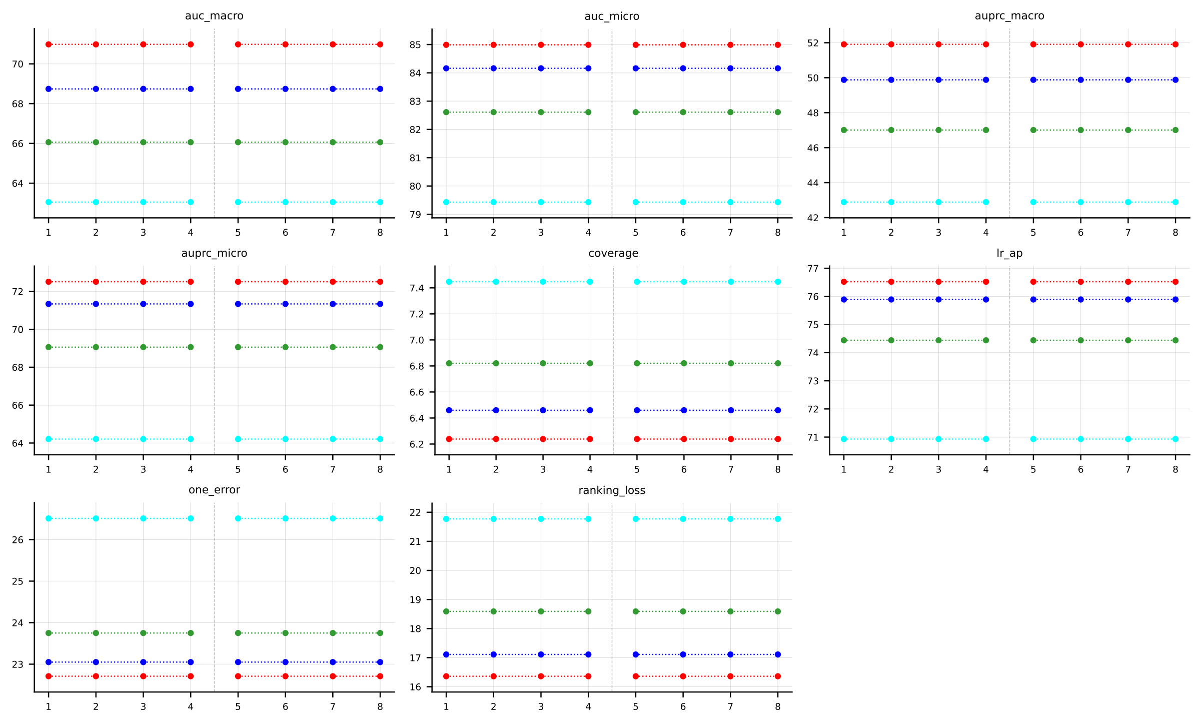

Table 22: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `Yeast` (RF base learner).

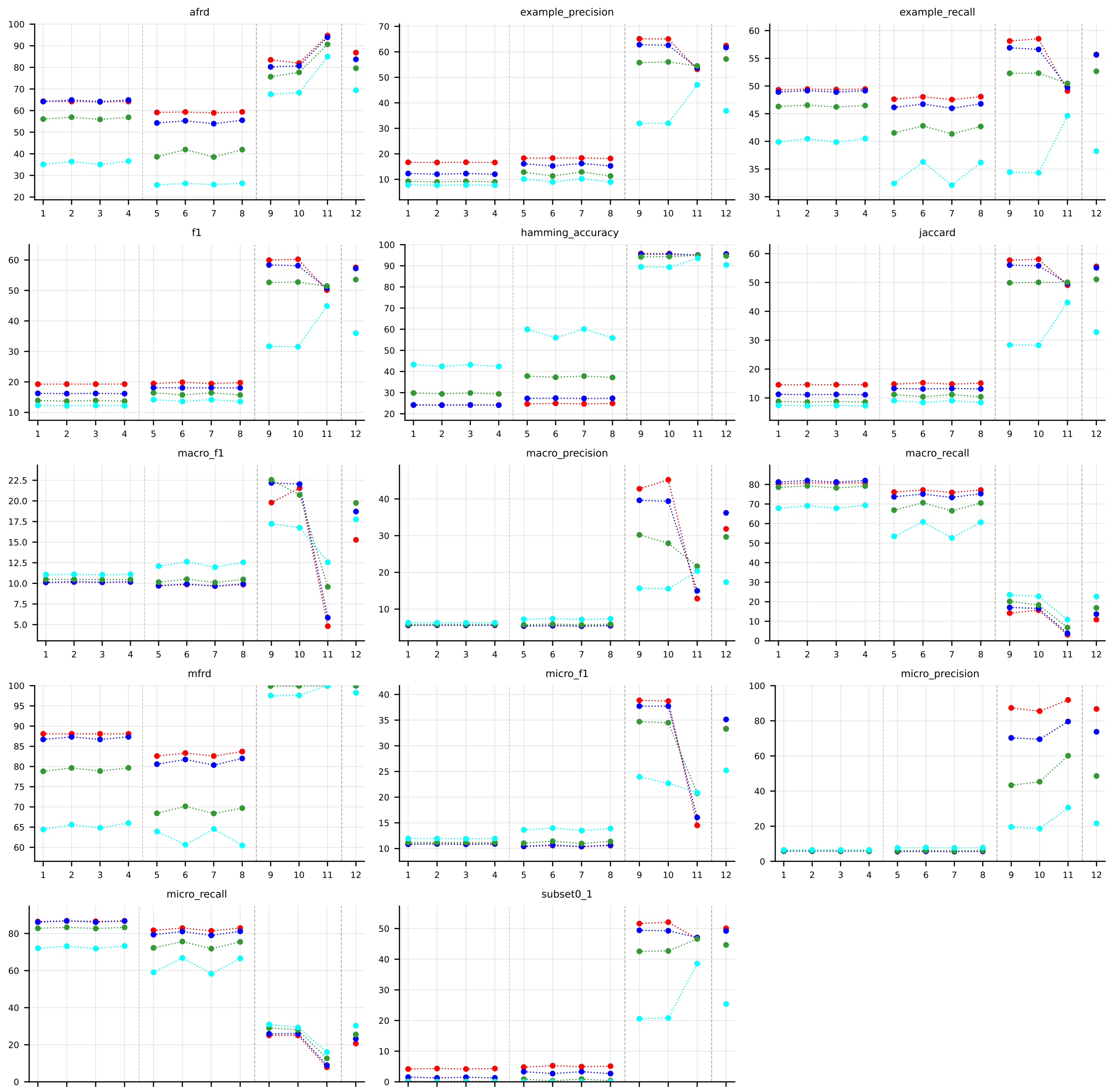

Table 23: Standard MLC predictions (BinaryVector) on `birds` (RF base learner).

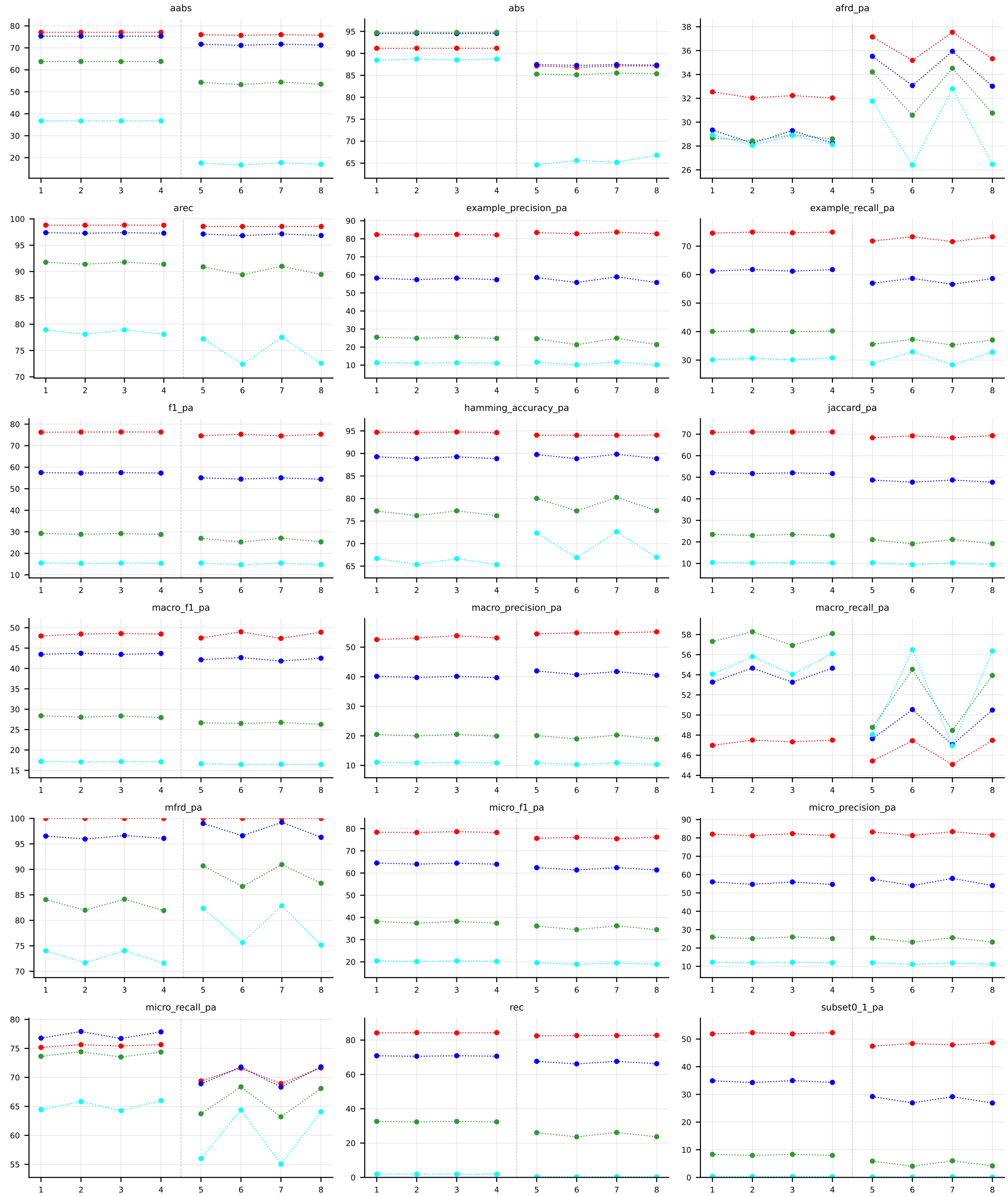

Table 24: Partial abstention (PartialAbstention) on `birds` (RF base learner).

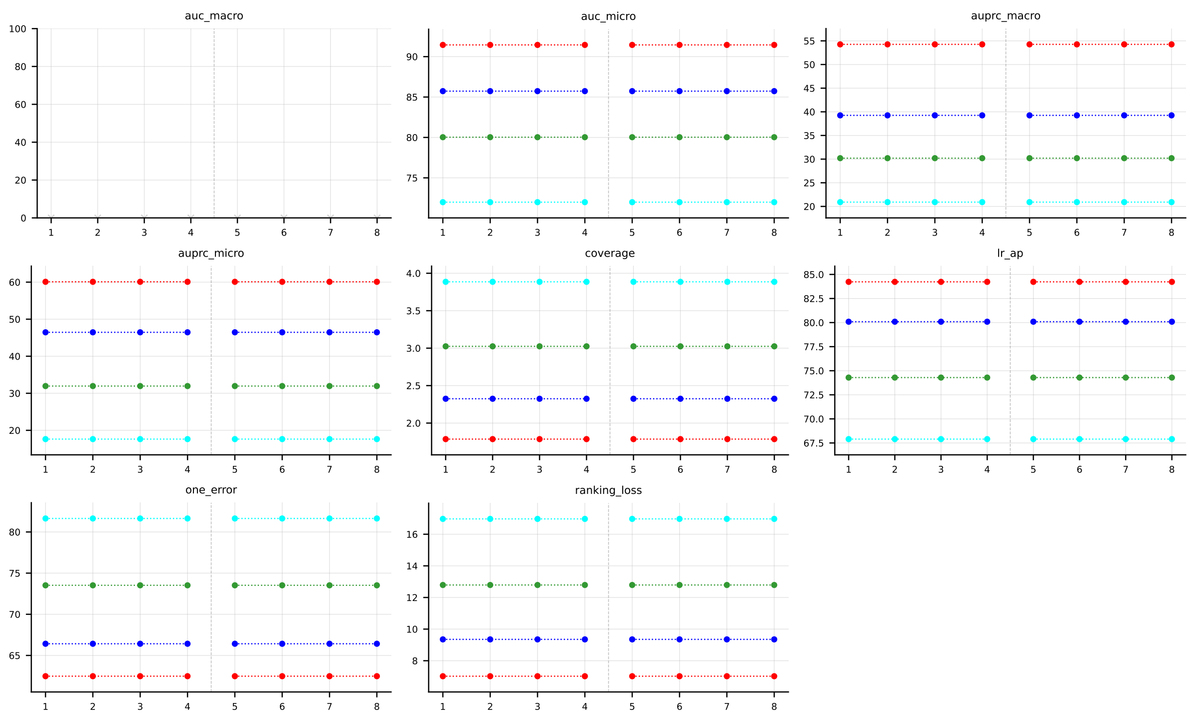

Table 25: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `birds` (RF base learner).

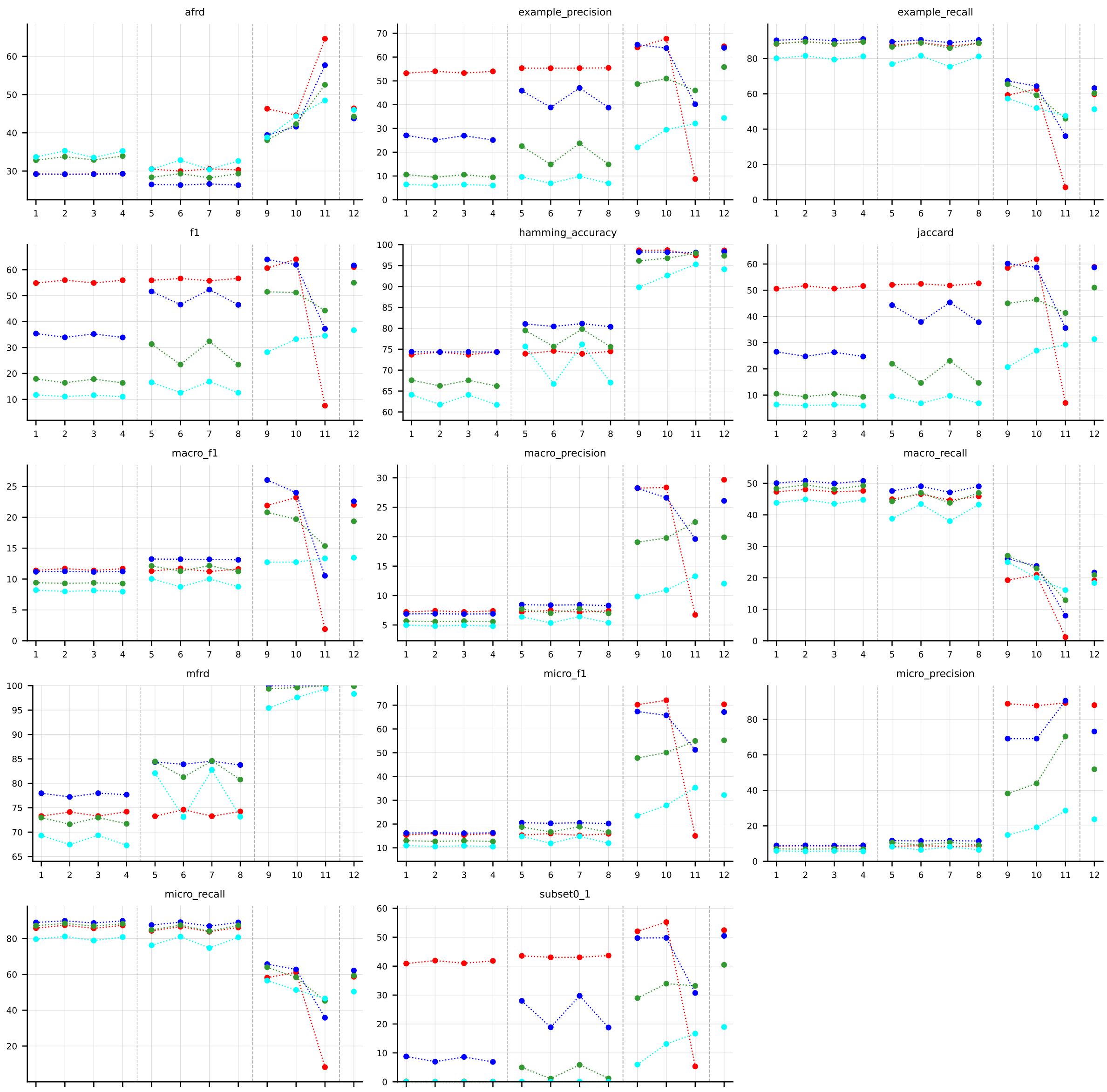

Table 26: Standard MLC predictions (BinaryVector) on `medical` (RF base learner).

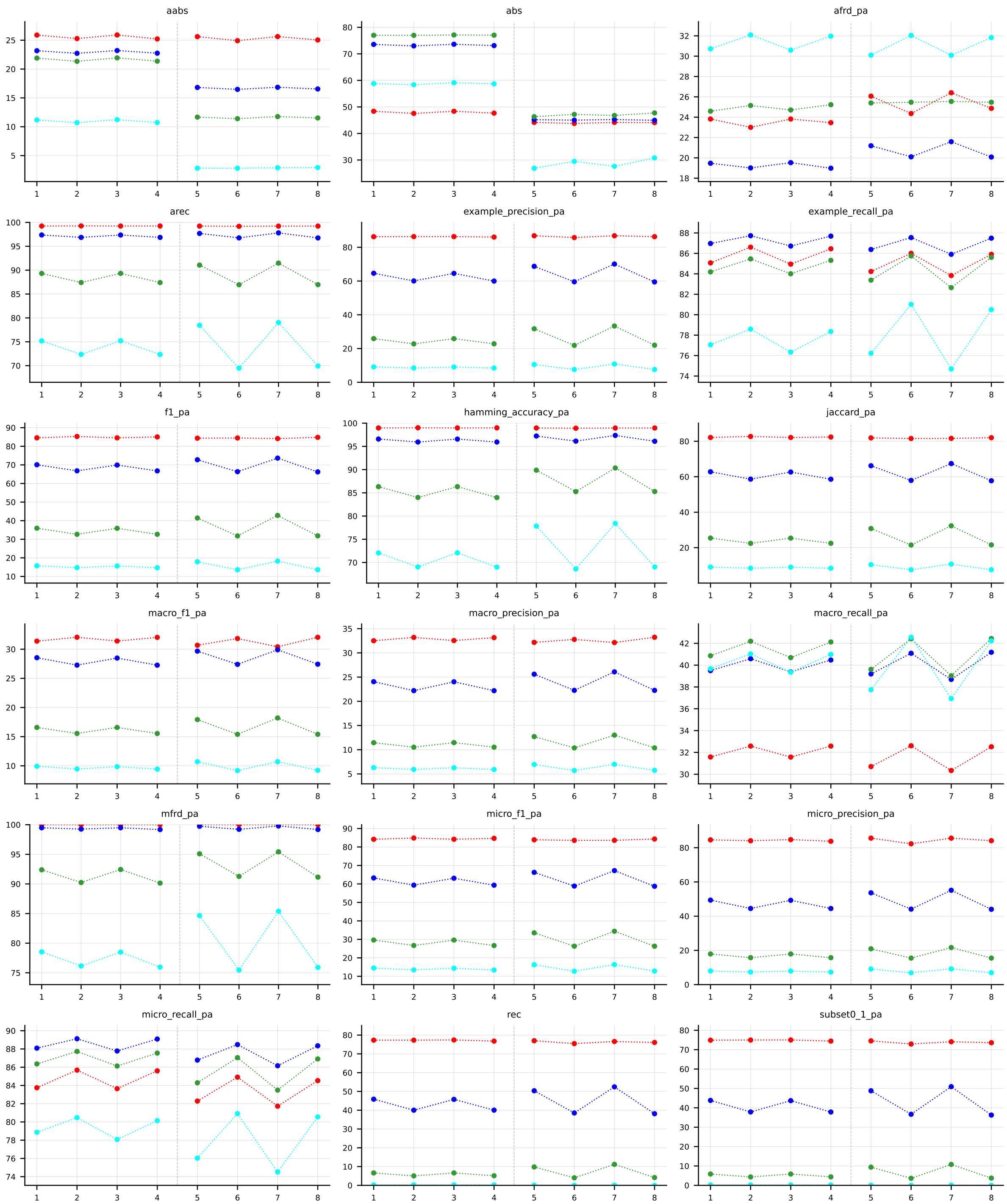

Table 27: Partial abstention (PartialAbstention) on `medical` (RF base learner).

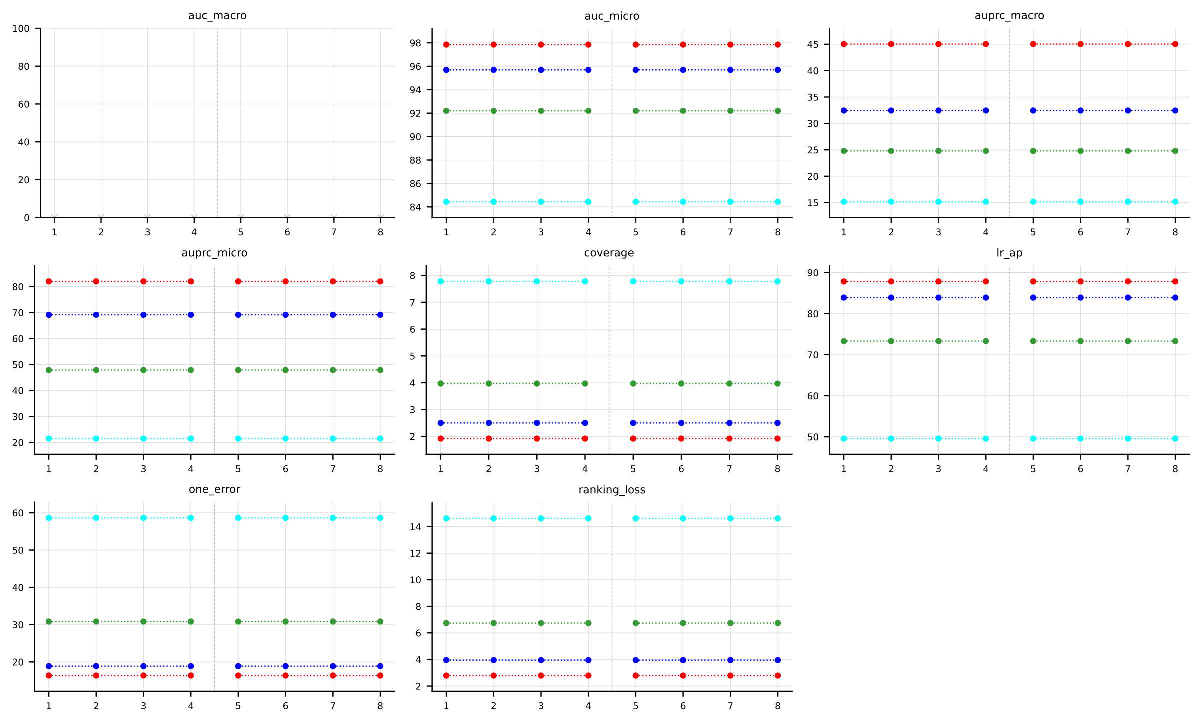

Table 28: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `medical` (RF base learner).

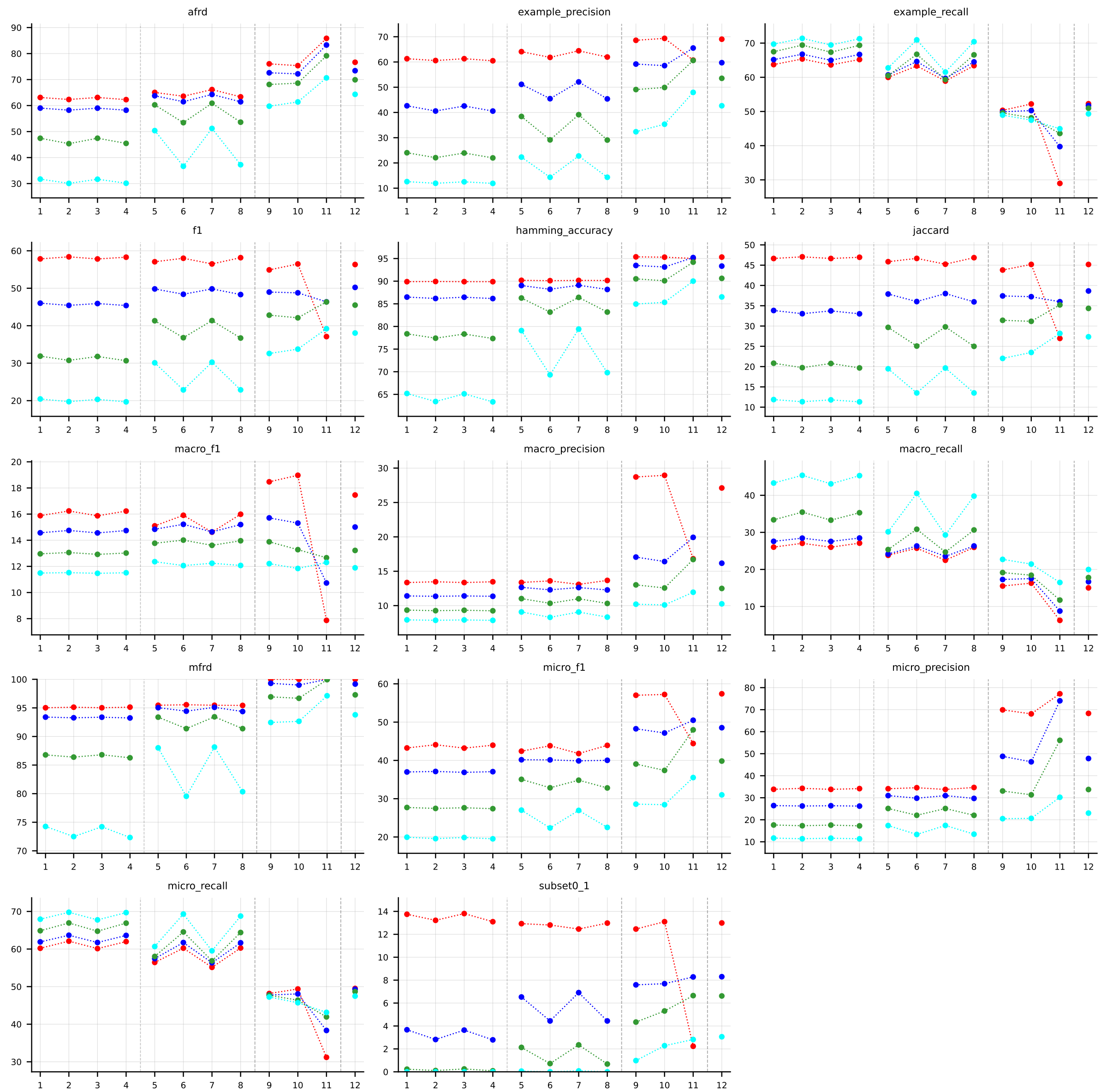

Table 29: Standard MLC predictions (BinaryVector) on `enron` (RF base learner).

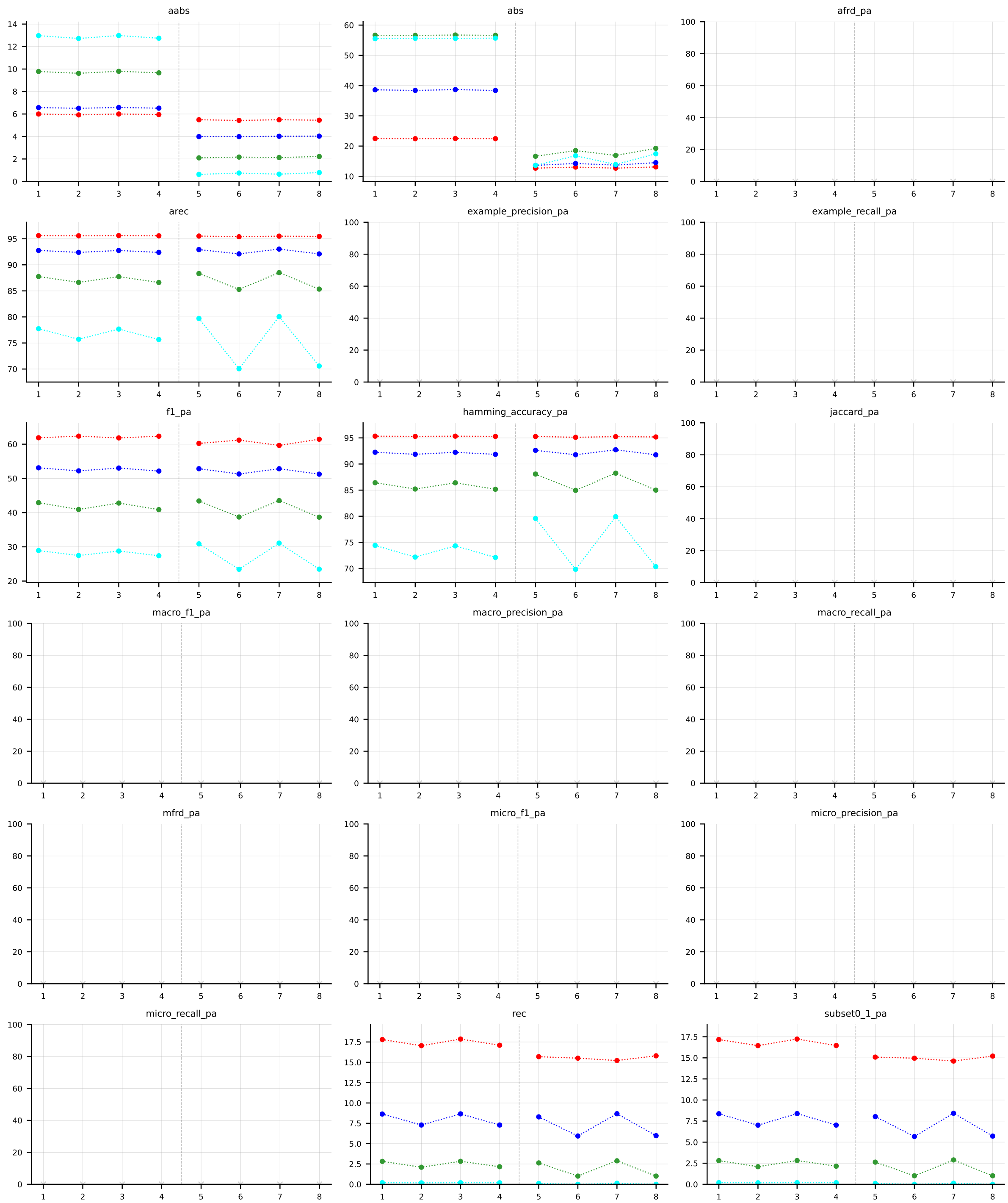

Table 30: Partial abstention (PartialAbstention) on `enron` (RF base learner).

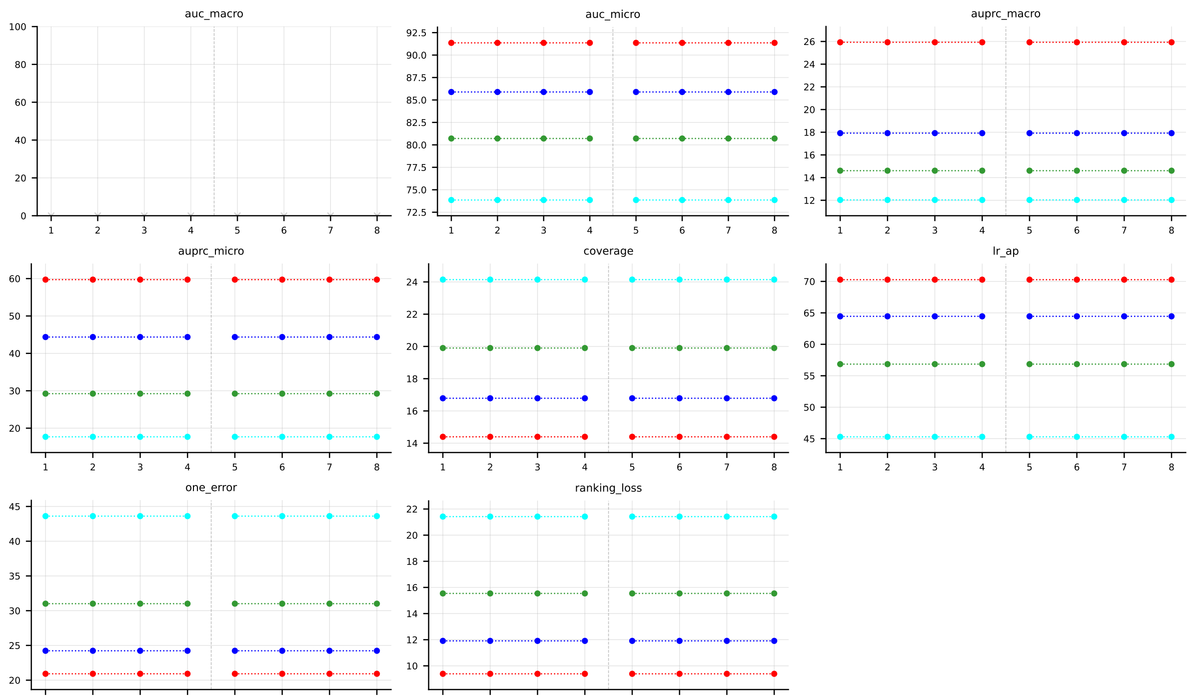

Table 31: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `enron` (RF base learner).

B LGBM aggregated results

LGBM • Aggregate • BinaryVector — Standard MLC predictions

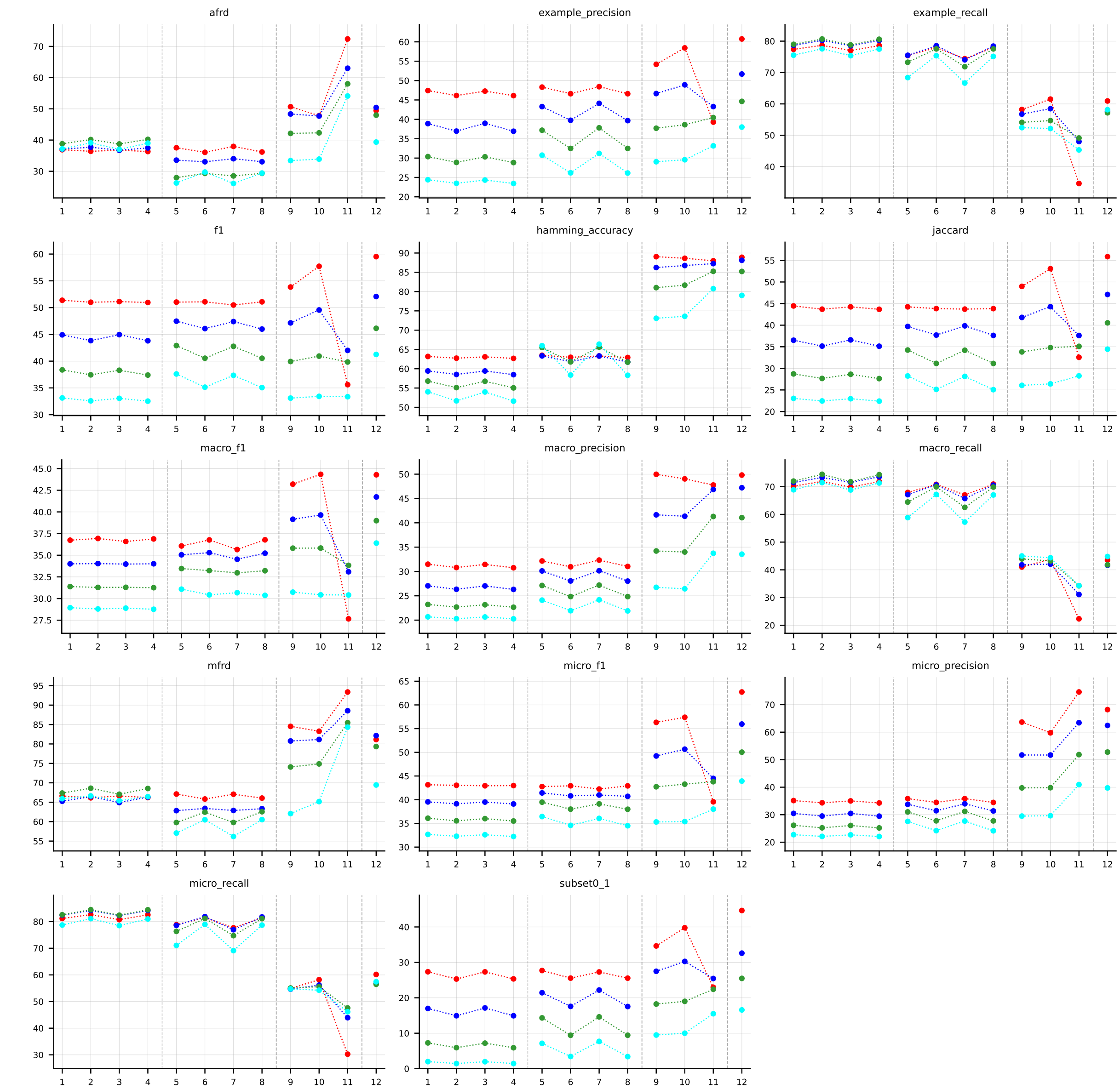

Table 32: Standard MLC predictions (BinaryVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

LGBM • Aggregate • PartialAbstention — Partial abstention

LGBM • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

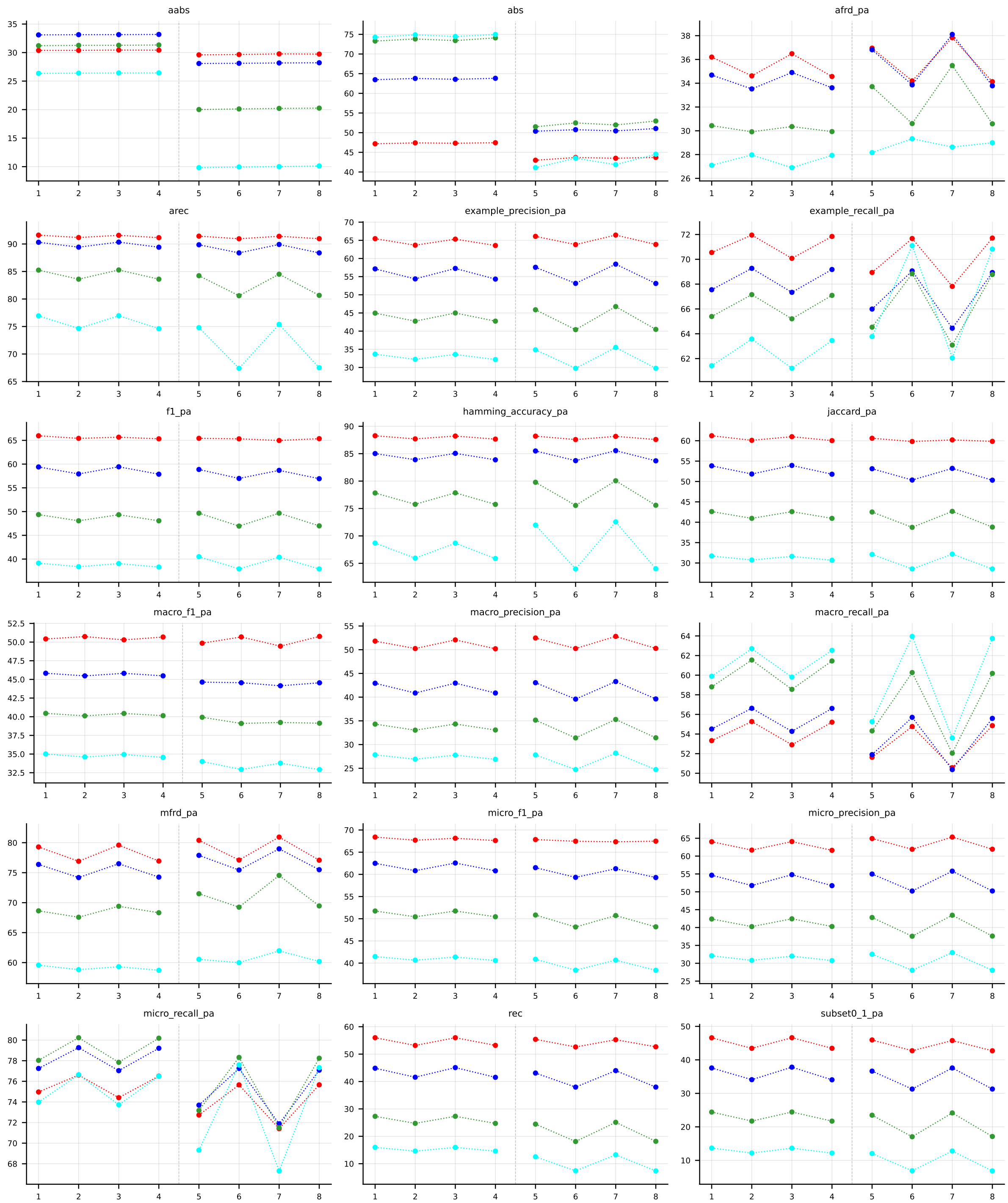

Table 33: Partial abstention (PartialAbstention), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

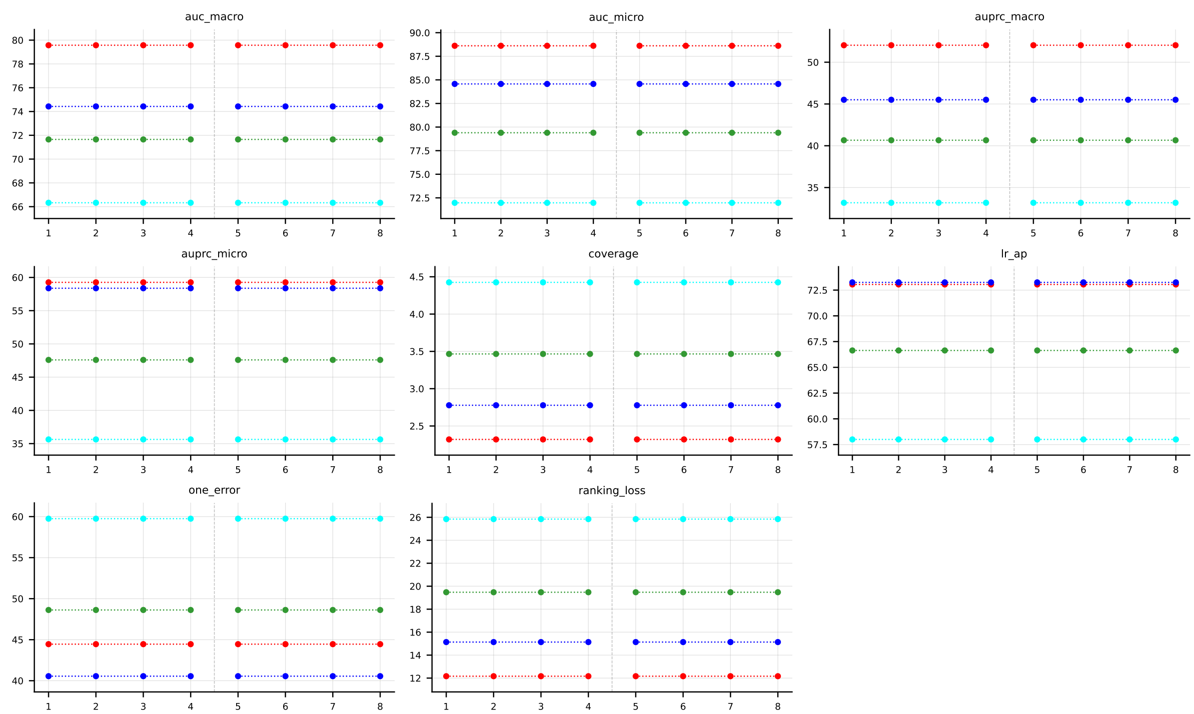

Table 34: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.